Multivariate Analysis of S&P 500 Sector Returns and a Review of Sparse Canonical Correlation Analysis

STAT Final Project

Yuan Gao, Tingzhou Wei, Yijie Huang

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1 Part I: Data Project — Multivariate Analysis of S&P 500 Sector Returns

1.1 Introduction

1.1.1 Motivation and Research Questions

The stock market provides a rich laboratory for multivariate statistical analysis. Individual securities and sector-level portfolios move together in complex, time-varying ways that reflect underlying macroeconomic forces, investor sentiment, and sector-specific dynamics. Understanding the covariance structure of returns across sectors is of both practical and theoretical interest: portfolio managers seek to exploit diversification benefits by holding assets whose returns are weakly correlated, while policymakers and academics use cross-sector linkages as a barometer of systemic risk.

In this project, we analyze the daily log-returns of eleven S&P 500 sector exchange-traded funds (ETFs) over the period January 2018 through December 2024, sourced via the `quantmod` package from Yahoo Finance. The eleven SPDR sector ETFs provide a clean, standardised partition of the S&P 500 universe and are widely used in both academic and industry research.

We address the following questions:

1. What are the dominant axes of variation in sector returns, and how much of total variance can be explained by a small number of latent factors? We apply **Principal Component Analysis (PCA)** to answer this.
2. Is there a significant linear relationship between *cyclical* sectors (Technology, Consumer Discretionary, Financials, Industrials) and *defensive* sectors (Healthcare, Utilities, Consumer Staples, Real Estate)? We apply **Canonical Correlation Analysis (CCA)** to answer this.
3. To what extent are the assumptions underlying these classical methods satisfied by financial return data, and what alternative methods might be more appropriate?

1.1.2 Related Literature

The factor structure of equity returns has been studied extensively since the Capital Asset Pricing Model (CAPM; Sharpe 1964) and the Arbitrage Pricing Theory (Ross 1976). More recently, Fama and French (1993, 2015) established that a small number of factors – market, size, value, profitability, and investment – explain much of the cross-sectional variation in returns. PCA applied to return covariance matrices has been used to extract such latent factors in a data-driven way (Connor and Korajczyk 1988; Bai and Ng 2002). CCA has been applied to study links between international equity markets (Bracker and Koch 1999) and between asset classes (Attié and Roache 2009). Our analysis follows this tradition but focuses on sector-level ETFs over a sample period that includes extreme episodes such as the COVID-19 market crash of 2020 and the 2022 rate-hiking cycle.

1.1.3 Data Overview

We use eleven SPDR Select Sector ETFs:

ETF	Sector	Classification
XLK	Technology	Cyclical
XLY	Consumer Discretionary	Cyclical
XLF	Financials	Cyclical
XLI	Industrials	Cyclical
XLB	Materials	Cyclical
XLV	Health Care	Defensive
XLP	Consumer Staples	Defensive
XLU	Utilities	Defensive
XLRE	Real Estate	Defensive
XLE	Energy	Other
XLC	Communication Services	Cyclical

Daily log-returns are computed as $r_t = \log(P_t/P_{t-1})$, where P_t is the adjusted closing price on day t. The sample runs from January 2, 2018 to December 31, 2024, yielding approximately 1,760 trading-day observations after removing holidays and missing data.

1.2 Data Collection and Preparation

Reproducibility is a recurring challenge when the primary data source is a free public API such as Yahoo Finance: connection errors, rate-limiting, and silent schema changes are common. To make this document reproducible we adopt a **three-tier strategy**:

1. **Local cache (fastest):** if the file `sector_returns_cache.rds` exists in the working directory, we load it directly and skip any downloads.
2. **Yahoo Finance (first-time setup):** otherwise we download each ETF individually using `quantmod::getSymbols()` inside a `tryCatch()` with retries; successful downloads are merged, cleaned, and saved both as an `.rds` file (fast binary) and a `.csv` file (human-readable backup).
3. **Failure diagnostics:** if any ticker still fails after retries, the code reports which one(s) so the user can manually supply a CSV.

```r
tickers <- c("XLK","XLY","XLF","XLI","XLB",
             "XLV","XLP","XLU","XLRE","XLE","XLC")

sector_names <- c("Technology","Cons.Disc","Financials","Industrials",
                  "Materials","HealthCare","Cons.Staples","Utilities",
                  "RealEstate","Energy","Comm.Serv")

start_date <- "2018-01-01"
end_date   <- "2024-12-31"

cache_rds <- "sector_returns_cache.rds"
cache_csv <- "sector_returns_cache.csv"

# ---- Tier 1: try local cache ------------------------------------------------
if (file.exists(cache_rds)) {
  log_returns <- readRDS(cache_rds)
  message("Loaded returns from local cache: ", cache_rds)
} else {

  # ---- Tier 2: download from Yahoo Finance, one ticker at a time ----------
  fetch_one <- function(tk, tries = 3, pause = 1) {
    for (k in seq_len(tries)) {
      res <- tryCatch(
        getSymbols(tk, src = "yahoo",
                   from = start_date, to = end_date,
                   auto.assign = FALSE, return.class = "xts"),
        error = function(e) e
      )
      if (!inherits(res, "error") && !is.null(res)) return(res)
      Sys.sleep(pause * k)   # simple exponential back-off
    }
```

```r
    NULL
  }

  prices_list <- lapply(tickers, fetch_one)
  names(prices_list) <- tickers

  failed <- tickers[sapply(prices_list, is.null)]
  if (length(failed) > 0) {
    stop("Failed to download the following ticker(s) after retries: ",
         paste(failed, collapse = ", "),
         "\nPlease supply a CSV manually (see instructions in the .Rmd).")
  }

  # Extract adjusted close for every ticker
  adj_list <- lapply(prices_list, Ad)
  prices   <- do.call(merge, adj_list)
  colnames(prices) <- tickers
  prices <- na.omit(prices)

  # Daily log-returns
  log_returns <- diff(log(prices))[-1, ]
  colnames(log_returns) <- sector_names

  # ---- Persist to disk so later knits are fast and offline-friendly -------
  saveRDS(log_returns, cache_rds)
  write.csv(data.frame(Date = index(log_returns),
                       coredata(log_returns)),
            cache_csv, row.names = FALSE)
  message("Saved cache to ", cache_rds, " and ", cache_csv)
}

cat("Sample period:", as.character(index(log_returns)[1]),
    "to", as.character(tail(index(log_returns), 1)), "\n")
```

```
## Sample period: 2018-06-20 to 2024-12-30
```

```r
cat("Number of trading days:", nrow(log_returns), "\n")
```

```
## Number of trading days: 1643
```

```r
cat("Number of sectors:",      ncol(log_returns), "\n")
```

```
## Number of sectors: 11
```

```r
# Convert to plain matrix for analysis
R <- as.matrix(log_returns)
```

```r
n <- nrow(R)
p <- ncol(R)
```

> **Note on the cache:** The first time you knit this document, an internet connection is required so that data can be pulled from Yahoo Finance. Thereafter the files `sector_returns_cache.rds` and `sector_returns_cache.csv` sit in your project folder and are used automatically — the document will knit offline. If you ever wish to refresh the data, simply delete those two files before knitting.

1.3 Exploratory Data Analysis

1.3.1 Summary Statistics

```
summ <- data.frame(
  Sector = sector_names,
  Mean   = round(colMeans(R) * 252, 4),           # annualised
  Std    = round(apply(R, 2, sd) * sqrt(252), 4), # annualised
  Skew   = round(apply(R, 2, function(x) mean((x - mean(x))^3) / sd(x)^3), 3),
  Kurt   = round(apply(R, 2, function(x) mean((x - mean(x))^4) / sd(x)^4), 3),
  Min    = round(apply(R, 2, min), 4),
  Max    = round(apply(R, 2, max), 4)
)

kable(summ, booktabs = TRUE, longtable = FALSE,
      caption = "Summary statistics for daily log-returns (mean and std are annualised).
                 Kurtosis values substantially above 3 indicate heavy tails.",
      col.names = c("Sector","Ann. Mean","Ann. Std","Skewness",
                    "Kurtosis","Min (daily)","Max (daily)")) %>%
  kable_styling(latex_options = c("hold_position","scale_down"), font_size = 9)
```

Table 1: Summary statistics for daily log-returns (mean and std are annualised). Kurtosis values substantially above 3 indicate heavy tails.

	Sector	Ann. Mean	Ann. Std	Skewness	Kurtosis	Min (daily)	Max (daily)
Technology	Technology	0.1925	0.2651	-0.428	10.841	-0.1487	0.1109
Cons.Disc	Cons.Disc	0.1177	0.2431	-0.744	10.520	-0.1355	0.0897
Financials	Financials	0.1070	0.2450	-0.565	18.207	-0.1474	0.1236
Industrials	Industrials	0.1078	0.2197	-0.587	16.029	-0.1204	0.1191
Materials	Materials	0.0756	0.2254	-0.493	12.673	-0.1166	0.1112
HealthCare	HealthCare	0.0907	0.1760	-0.394	14.314	-0.1038	0.0742
Cons.Staples	Cons.Staples	0.0926	0.1592	-0.487	19.631	-0.0987	0.0817
Utilities	Utilities	0.0950	0.2134	-0.226	18.391	-0.1206	0.1204
RealEstate	RealEstate	0.0720	0.2314	-1.086	20.451	-0.1744	0.0842
Energy	Energy	0.0652	0.3360	-0.909	17.929	-0.2249	0.1487
Comm.Serv	Comm.Serv	0.1112	0.2318	-0.554	9.580	-0.1197	0.0861

Several features stand out. First, all sectors exhibit negative skewness, consistent with the well-documented negative skewness of equity returns attributable to crash risk. Second, kurtosis values are substantially larger than 3 (the Gaussian value) for all sectors, confirming heavy-tailed behaviour. Energy (XLE) shows the highest volatility and most extreme kurtosis. Technology and Communication Services exhibit the highest annualised means over this sample, driven partly by the 2023–2024 AI-driven rally.

1.3.2 Return Distributions

```r
R_long <- as.data.frame(R) %>%
  mutate(Date = index(log_returns)) %>%
  pivot_longer(-Date, names_to = "Sector", values_to = "Return")

# Per-sector normal fit (correct overlay; the previous version used a
# single global mean/sd for every facet, which is misleading).
norm_curves <- R_long %>%
  group_by(Sector) %>%
  summarise(m = mean(Return), s = sd(Return),
            lo = min(Return), hi = max(Return), .groups = "drop") %>%
  rowwise() %>%
  do({
    x <- seq(.$lo, .$hi, length.out = 200)
    data.frame(Sector = .$Sector, Return = x,
               NormDens = dnorm(x, mean = .$m, sd = .$s))
  }) %>% ungroup()

ggplot(R_long, aes(x = Return)) +
  geom_density(fill = "steelblue", alpha = 0.4, color = "steelblue3") +
  geom_line(data = norm_curves, aes(x = Return, y = NormDens),
            color = "red", linetype = "dashed", linewidth = 0.6) +
  facet_wrap(~ Sector, scales = "free", ncol = 4) +
  labs(x = "Daily Log Return", y = "Density") +
  theme_bw(base_size = 9) +
  theme(strip.background = element_rect(fill = "grey90"))
```

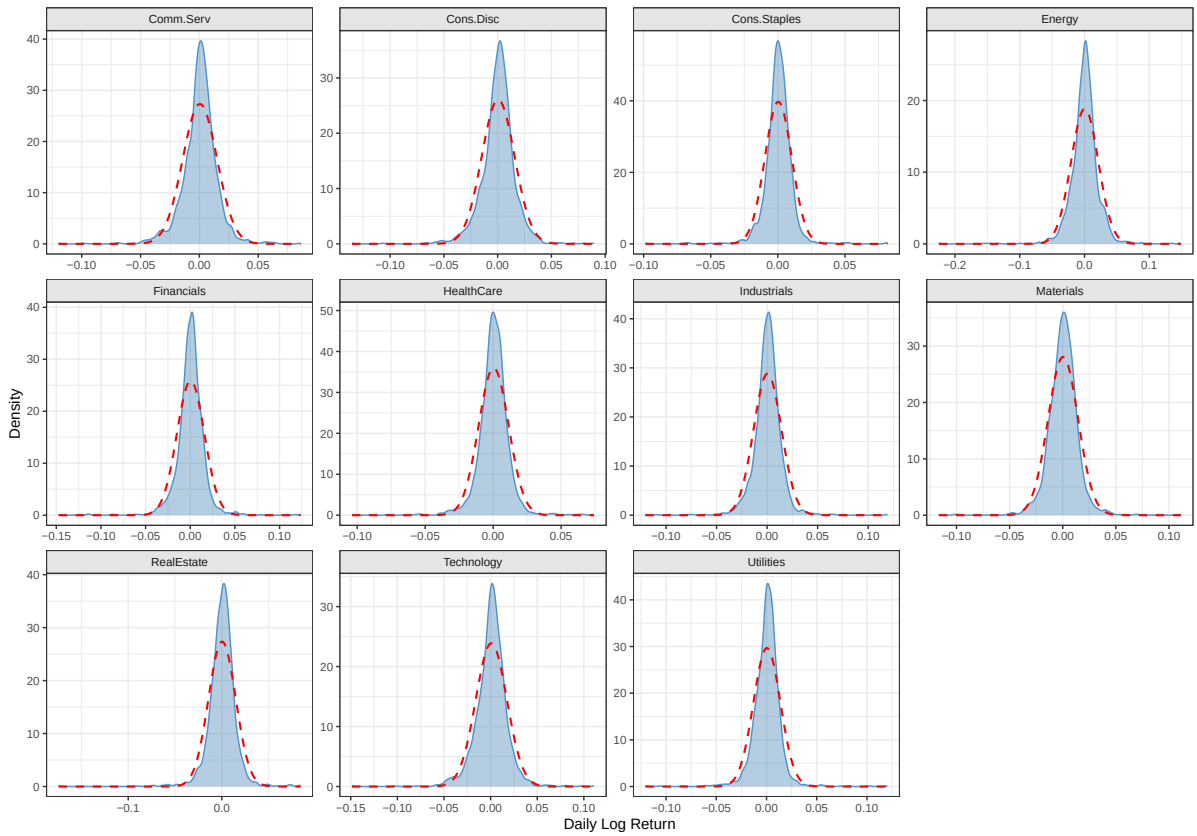

Figure 1: Kernel density estimates of daily log-returns for each sector (blue) overlaid with the fitted *sector-specific* normal density (red dashed). Heavier tails and sharper peaks than the normal are visible in all panels.

1.3.3 Correlation Structure

```
corr_mat <- cor(R)
rownames(corr_mat) <- colnames(corr_mat) <- sector_names

# Reorder: cyclicals (incl. Comm.Serv), then defensives, then Energy last
ord <- c(1,2,3,4,5,11,  6,7,8,9,  10)
corr_ord <- corr_mat[ord, ord]

ggcorrplot(corr_ord, method = "square", type = "full",
           lab = TRUE, lab_size = 2.8,
           colors = c("#D73027","white","#4575B4"),
           title  = "Sector Return Correlation Matrix",
           ggtheme = theme_bw()) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
        axis.text.y = element_text(size = 8))
```

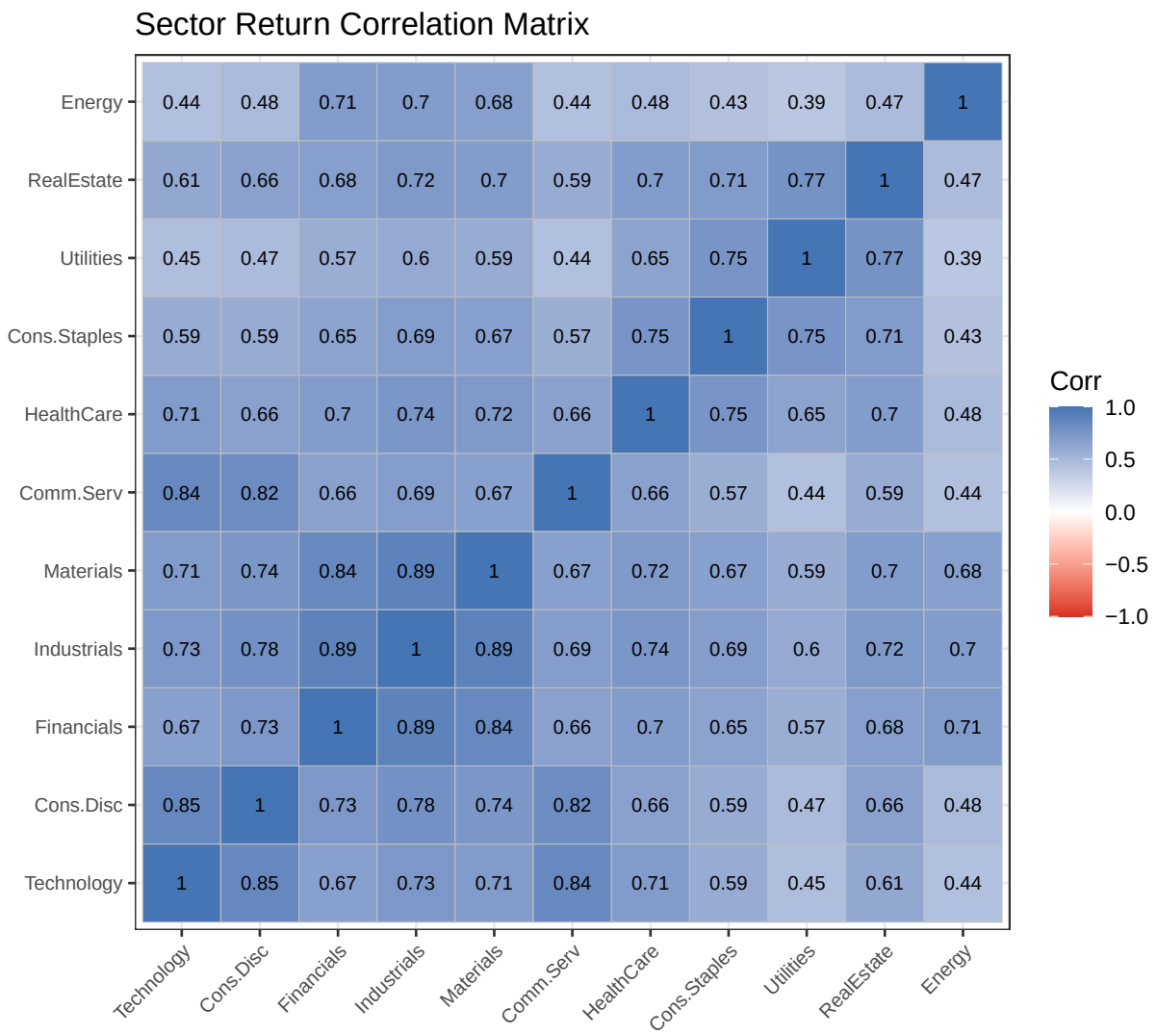

Figure 2: Pearson correlation matrix of daily log-returns. Cyclical sectors (top-left block) show stronger mutual correlations than defensive sectors. The Energy sector stands apart from both groups.

The correlation matrix reveals a clear block structure. The six cyclical sectors (Technology, Consumer Discretionary, Financials, Industrials, Materials, Communication Services) have pairwise correlations generally in the range 0.6–0.85. The four defensive sectors (Health Care, Consumer Staples, Utilities, Real Estate) are somewhat less correlated with each other (0.4–0.65) and substantially less correlated with cyclicals. Energy occupies an intermediate position, exhibiting moderate correlations with all other sectors. This block structure motivates the CCA in Section 5, where we formally test and quantify the linear relationship between the cyclical and defensive groups.

1.3.4 Normality Tests

The visual evidence of non-normality is corroborated by three complementary diagnostics: Mardia's classical moment-based test, the energy goodness-of-fit test of Sz'ekely and Rizzo (2005), and a χ^2_p Q–Q plot of squared Mahalanobis distances. The three approaches are mutually informative: Mardia targets multivariate skewness and kurtosis through fixed moments; the energy test is consistent against *any* departure

from multivariate normality (not only moment misspecification); and the Q–Q plot localises *where* in the distribution the deviation occurs.

```r
library(energy)

# ---- (a) Mardia's test (manual; avoids the fragile MVN package) ----------
mardia_test <- function(X) {
  X <- as.matrix(X); n <- nrow(X); p <- ncol(X)
  Xc <- scale(X, scale = FALSE)
  S  <- crossprod(Xc) / n
  D  <- Xc %*% solve(S) %*% t(Xc)

  b1p <- sum(D^3) / n^2                # multivariate skewness
  b2p <- sum(diag(D)^2) / n            # multivariate kurtosis

  skew_stat <- n * b1p / 6
  skew_df   <- p * (p + 1) * (p + 2) / 6
  skew_p    <- pchisq(skew_stat, skew_df, lower.tail = FALSE)

  kurt_z <- (b2p - p * (p + 2)) / sqrt(8 * p * (p + 2) / n)
  kurt_p <- 2 * pnorm(-abs(kurt_z))

  list(skew = c(stat = skew_stat, df = skew_df, p = skew_p),
       kurt = c(stat = kurt_z,   df = NA,      p = kurt_p))
}

mar <- mardia_test(R)

# ---- (b) Energy GOF test for MVN (Szekely-Rizzo 2005) --------------------
# Distribution-free under H1; consistent against all departures from MVN.
# Critical values via parametric-bootstrap calibration (R replicates).
set.seed(2024)
en <- energy::mvnorm.test(R, R = 4999)

# ---- Combined table -------------------------------------------------------
norm_tbl <- data.frame(
  Test = c("Mardia skewness ($\\chi^2$)",
           "Mardia kurtosis ($Z$)",
           "Energy GOF ($n\\,\\mathcal{E}_{n,d}$)"),
  Statistic = c(round(mar$skew["stat"], 2),
                round(mar$kurt["stat"], 2),
                round(unname(en$statistic), 2)),
  df = c(mar$skew["df"], NA, NA),
  p.value = c(format.pval(mar$skew["p"], digits = 3, eps = .Machine$double.eps),
```

```r
              format.pval(mar$kurt["p"], digits = 3, eps = .Machine$double.eps),
              format.pval(en$p.value,    digits = 3, eps = 1/5000)),
  stringsAsFactors = FALSE
)

kable(norm_tbl, booktabs = TRUE, escape = FALSE,
      caption = "Three tests of multivariate normality on the eleven-sector
                 daily log-return panel. All three reject MVN decisively. The
                 energy test $p$-value is bounded below by $1/(R+1) = 1/5000$
                 by the permutation calibration.",
      col.names = c("Test", "Statistic", "df", "$p$-value")) %>%
  kable_styling(latex_options = "hold_position")
```

Table 2: Three tests of multivariate normality on the eleven-sector daily log-return panel. All three reject MVN decisively. The energy test p-value is bounded below by $1/(R+1) = 1/5000$ by the permutation calibration.

Test	Statistic	df	p-value
Mardia skewness (χ^2)	2605.64	286	<2e-16
Mardia kurtosis (Z)	161.53	NA	<2e-16
Energy GOF ($n\,\mathcal{E}_{n,d}$)	23.04	NA	<2e-04

```r
n_obs <- nrow(R); p_obs <- ncol(R)
Xc <- scale(R, scale = FALSE)
S  <- crossprod(Xc) / (n_obs - 1)
D2 <- as.numeric(rowSums((Xc %*% solve(S)) * Xc))
qx <- qchisq(ppoints(n_obs), df = p_obs)

qq_df <- data.frame(
  theo = sort(qx),
  emp  = sort(D2),
  date = index(log_returns)[order(D2)]
)
top3 <- tail(qq_df, 3)

ggplot(qq_df, aes(theo, emp)) +
  geom_point(alpha = 0.4, size = 0.9, color = "steelblue") +
  geom_abline(slope = 1, intercept = 0, color = "firebrick", linewidth = 0.8) +
  scale_x_log10() + scale_y_log10() +
  geom_text(data = top3,
            aes(label = format(date, "%Y-%m-%d")),
            hjust = 1.05, vjust = 0.5, size = 3) +
  labs(x = expression(paste("Theoretical ", chi[p]^2, " quantile (log scale)")),
```

```
      y = expression(paste("Sample ", D^2, " (log scale)"))) +
  theme_bw()
```

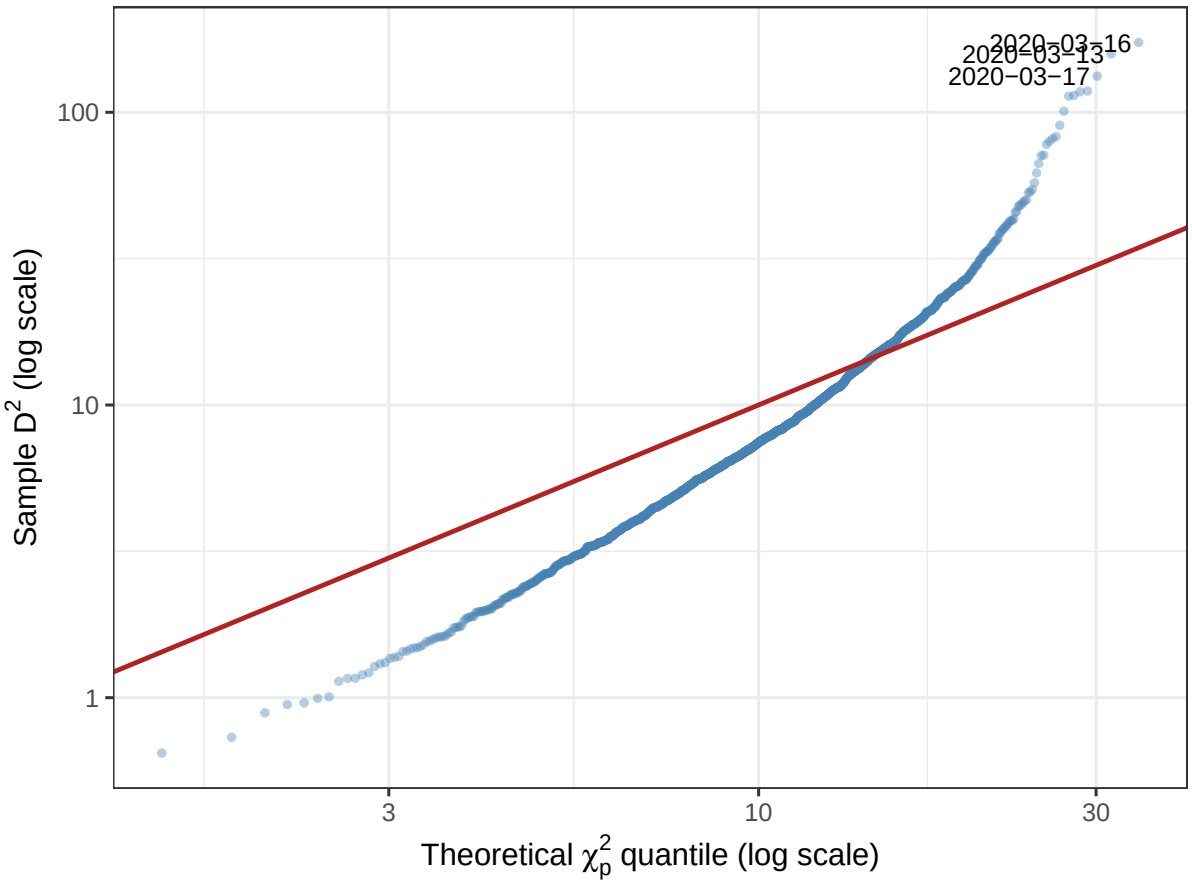

Figure 3: χ^2_p Q–Q plot of squared Mahalanobis distances $D_i^2 = (\mathbf{r}_i - \bar{\mathbf{r}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{r}_i - \bar{\mathbf{r}})$ for the eleven-sector return vector. Under multivariate normality the points should lie on the 45-degree reference line. The pronounced upward deviation in the right tail confirms that extreme co-movements occur far more frequently than a Gaussian model predicts; the three most extreme days are labelled.

```
# Diagnostic for the prose immediately below
exceed_99 <- mean(D2 > qchisq(0.99, p_obs))
```

The three diagnostics agree: sector returns are emphatically not multivariate normal. Mardia's skewness and kurtosis statistics yield p-values indistinguishable from zero in double precision; the energy goodness-of-fit test rejects at the smallest p-value the permutation calibration can deliver ($p < 1/5000$); and the χ^2_p Q–Q plot reveals the source of the rejection. The empirical fraction of days with $D_i^2 > \chi^2_{p,0.99}$ is approximately 6.5%, six to seven times the nominal 1% that an MVN would predict, and the upper-tail quantiles are far heavier than the χ^2_p reference. The most extreme observations correspond to known crisis days, indicating that the heavy tails reflect episodes of synchronised market stress rather than a handful of isolated outliers.

This pattern of *joint* extreme co-movement is precisely the regime in which the asymptotic Wilks' Λ test for canonical correlations breaks down; we therefore complement the asymptotic test with a distribution-free permutation analogue in the CCA section that follows.

1.3.5 Time Series Structure

```
cum_ret <- apply(R, 2, cumsum)
cum_df  <- as.data.frame(cum_ret) %>%
  mutate(Date = index(log_returns)) %>%
  pivot_longer(-Date, names_to = "Sector", values_to = "CumReturn")

ggplot(cum_df, aes(x = Date, y = CumReturn, color = Sector)) +
  geom_line(linewidth = 0.6) +
  scale_color_viridis_d(option = "turbo") +
  labs(x = NULL, y = "Cumulative Log Return", color = "Sector") +
  theme_bw() +
  theme(legend.position = "right", legend.text = element_text(size = 8))
```

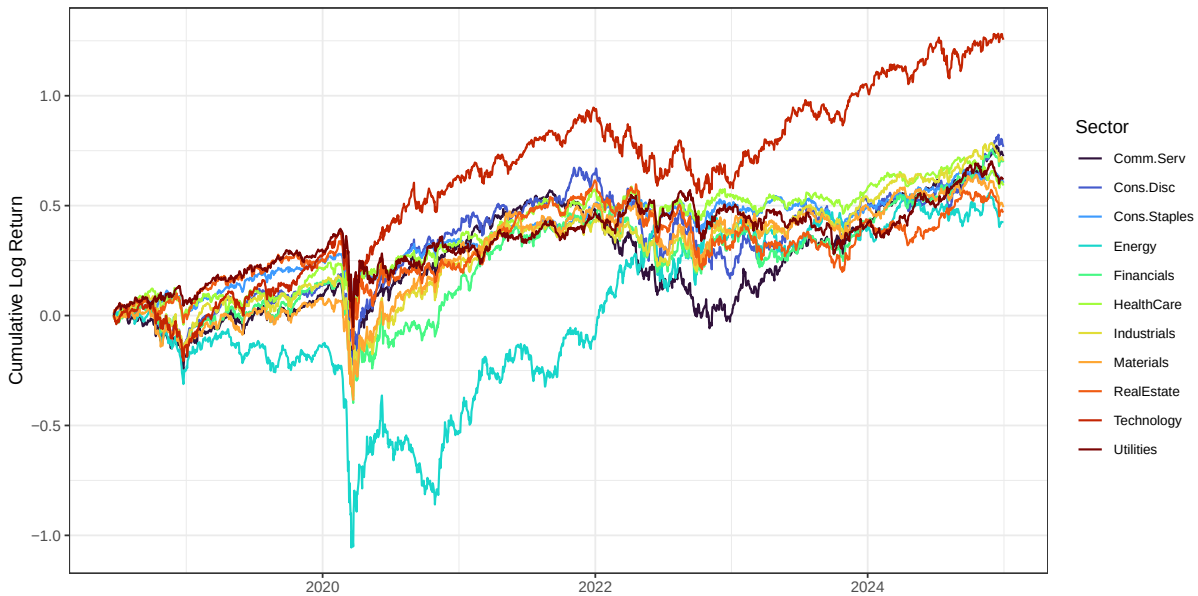

Figure 4: Cumulative log-return paths for all eleven sectors, 2018–2024. The COVID-19 crash (March 2020) and the 2022 rate cycle are clearly visible.

```
# NOTE: previous version mistakenly used tickers (XLK, XLV, ...) here even
# though `log_returns` carries sector_names. Fixed below and `rollapply` is
# computed only once.
roll_sd_xts <- rollapply(log_returns, width = 60, FUN = sd, by.column = TRUE)
roll_sd_xts <- na.omit(roll_sd_xts)

roll_sd <- data.frame(Date = index(roll_sd_xts),
                      coredata(roll_sd_xts),
                      check.names = FALSE) %>%
  select(Date, Technology, HealthCare, Energy, Utilities)
```

```r
roll_long <- pivot_longer(roll_sd, -Date,
                          names_to = "Sector", values_to = "RollSD")

ggplot(roll_long, aes(x = Date, y = RollSD * sqrt(252), color = Sector)) +
  geom_line(linewidth = 0.7) +
  scale_color_manual(values = c("steelblue","firebrick",
                                "goldenrod3","seagreen4")) +
  labs(x = NULL, y = "Annualised Rolling Volatility (60-day)") +
  theme_bw()
```

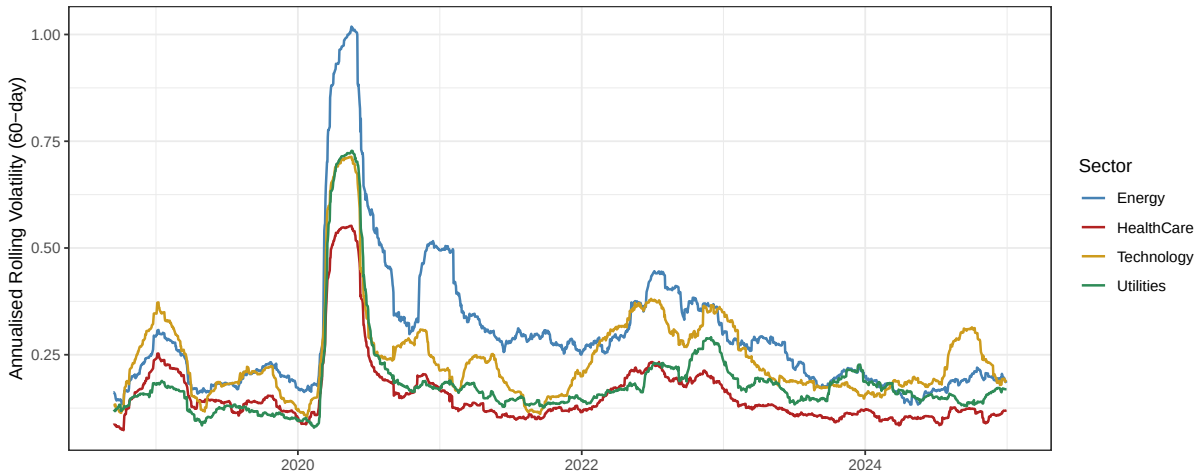

Figure 5: 60-day rolling standard deviation (annualised) for selected sectors. Volatility clustering is pronounced, with spikes during the March 2020 COVID crash.

The rolling volatility plot confirms substantial time variation in variance — a signature of GARCH-type dynamics. This violates the i.i.d. assumption underlying classical PCA and CCA inference. We address this further in the Discussion.

1.4 Principal Component Analysis

1.4.1 Motivation and Setup

PCA seeks an orthogonal linear transformation of the $n \times p$ data matrix $\mathbf{R}$ such that the projected coordinates have successively maximal variance. Let $\mathbf{\Sigma}$ denote the population covariance matrix of returns. The k-th principal component direction $\mathbf{v}_k$ solves

$$\mathbf{v}_k = \arg \max_{\|\mathbf{v}\|=1,\, \mathbf{v} \perp \mathbf{v}_1, \ldots, \mathbf{v}_{k-1}} \mathbf{v}^\top \mathbf{\Sigma} \mathbf{v},$$

with solution given by the k-th eigenvector of $\mathbf{\Sigma}$. In practice we substitute the sample covariance matrix $\hat{\mathbf{\Sigma}} = \frac{1}{n-1}\mathbf{R}^\top \mathbf{R}$ (with mean-centred columns). The k-th principal component score is $\mathbf{z}_k = \mathbf{R}\mathbf{v}_k$, and the proportion of variance explained by the first K components is $\sum_{k=1}^{K} \lambda_k / \sum_{k=1}^{p} \lambda_k$, where λ_k are the ordered eigenvalues.

We perform PCA on the demeaned return matrix. Because all variables are measured in the same units (log-returns), we use the covariance matrix rather than the correlation matrix, which would give equal weight to each sector regardless of its volatility.

1.4.2 Scree Plot and Variance Explained

```r
# Mean-centre
R_c <- scale(R, center = TRUE, scale = FALSE)

pca_res <- prcomp(R_c, center = FALSE, scale. = FALSE)
pve     <- pca_res$sdev^2 / sum(pca_res$sdev^2)
cpve    <- cumsum(pve)
```

```r
pve_df <- data.frame(PC = 1:p, PVE = pve * 100, CPVE = cpve * 100)

p1 <- ggplot(pve_df, aes(x = PC, y = PVE)) +
  geom_col(fill = "steelblue", alpha = 0.8) +
  geom_line(aes(group = 1), color = "firebrick", linewidth = 0.8) +
  geom_point(color = "firebrick", size = 2) +
  scale_x_continuous(breaks = 1:p) +
  labs(x = "Principal Component", y = "PVE (%)") +
  theme_bw()

p2 <- ggplot(pve_df, aes(x = PC, y = CPVE)) +
  geom_line(color = "steelblue3", linewidth = 1) +
  geom_point(color = "steelblue3", size = 2) +
  geom_hline(yintercept = 80, linetype = "dashed", color = "firebrick") +
  geom_hline(yintercept = 90, linetype = "dotted", color = "grey50") +
  annotate("text", x = p - 2, y = 82, label = "80%",
           color = "firebrick", size = 3) +
```

```r
  scale_x_continuous(breaks = 1:p) +
  labs(x = "Principal Component", y = "Cumulative PVE (%)") +
  theme_bw()

grid.arrange(p1, p2, ncol = 2)
```

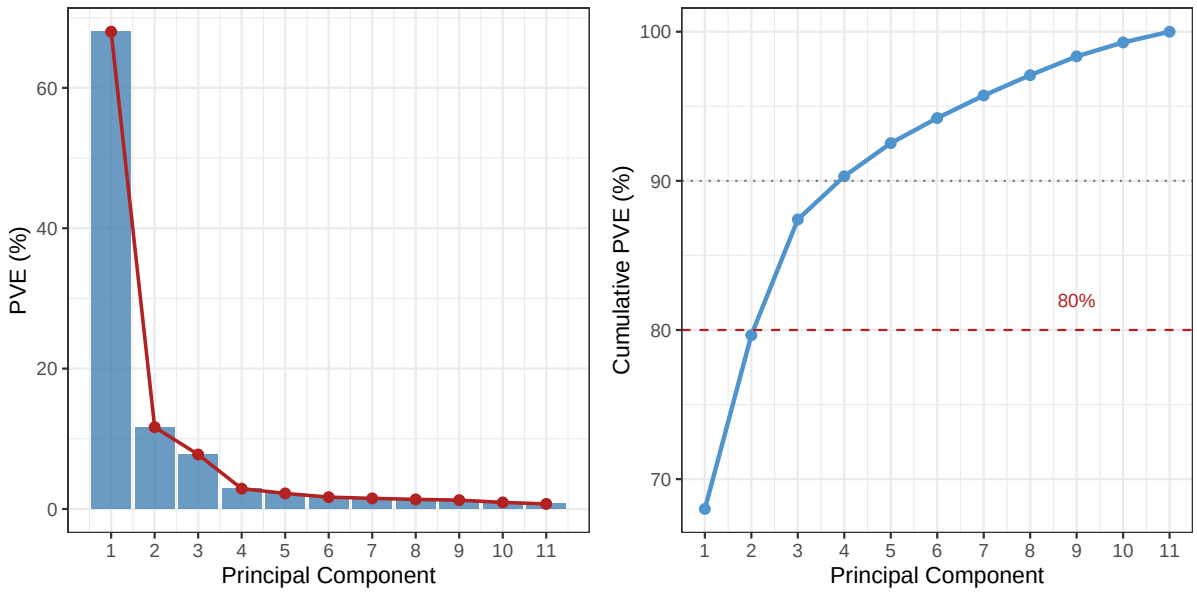

Figure 6: Left: Scree plot of proportion of variance explained (PVE) by each PC. Right: Cumulative PVE. The elbow at PC2 and the 80% threshold at PC4 suggest that four factors capture most of the systematic variation.

```r
pve_tbl <- data.frame(
  PC          = paste0("PC", 1:p),
  Eigenvalue = round(pca_res$sdev^2 * (n - 1), 4),
  PVE         = paste0(round(pve  * 100, 1), "%"),
  CPVE        = paste0(round(cpve * 100, 1), "%")
)
kable(pve_tbl, booktabs = TRUE,
      caption = "Eigenvalues and variance explained for each principal component.",
      col.names = c("Component","Eigenvalue","PVE","Cumulative PVE")) %>%
  kable_styling(latex_options = "hold_position")
```

Table 3: Eigenvalues and variance explained for each principal component.

Component	Eigenvalue	PVE	Cumulative PVE
PC1	2.7047	68%	68%
PC2	0.4635	11.7%	79.7%
PC3	0.3087	7.8%	87.4%
PC4	0.1152	2.9%	90.3%

PC5	0.0882	2.2%	92.5%
PC6	0.0670	1.7%	94.2%
PC7	0.0601	1.5%	95.7%
PC8	0.0543	1.4%	97.1%
PC9	0.0501	1.3%	98.3%
PC10	0.0372	0.9%	99.3%
PC11	0.0286	0.7%	100%

The first principal component alone explains approximately 68% of total variance. This dominant factor is interpreted below as the market-wide factor, analogous to the market return in the CAPM. The first four components together explain 90.3% of variance, consistent with the finding in the asset pricing literature that equity returns have a low-rank factor structure.

1.4.3 Loadings Interpretation

```r
load_mat <- pca_res$rotation[, 1:4]
rownames(load_mat) <- sector_names
colnames(load_mat) <- paste0("PC", 1:4)

load_df <- as.data.frame(load_mat) %>%
  mutate(Sector = sector_names) %>%
  pivot_longer(-Sector, names_to = "PC", values_to = "Loading")

ggplot(load_df, aes(x = PC, y = Sector, fill = Loading)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Loading, 2)), size = 3) +
  scale_fill_gradient2(low = "#D73027", mid = "white", high = "#4575B4",
                       midpoint = 0, limits = c(-0.7, 0.7)) +
  labs(x = NULL, y = NULL, fill = "Loading") +
  theme_bw() +
  theme(axis.text.x = element_text(size = 10, face = "bold"))
```

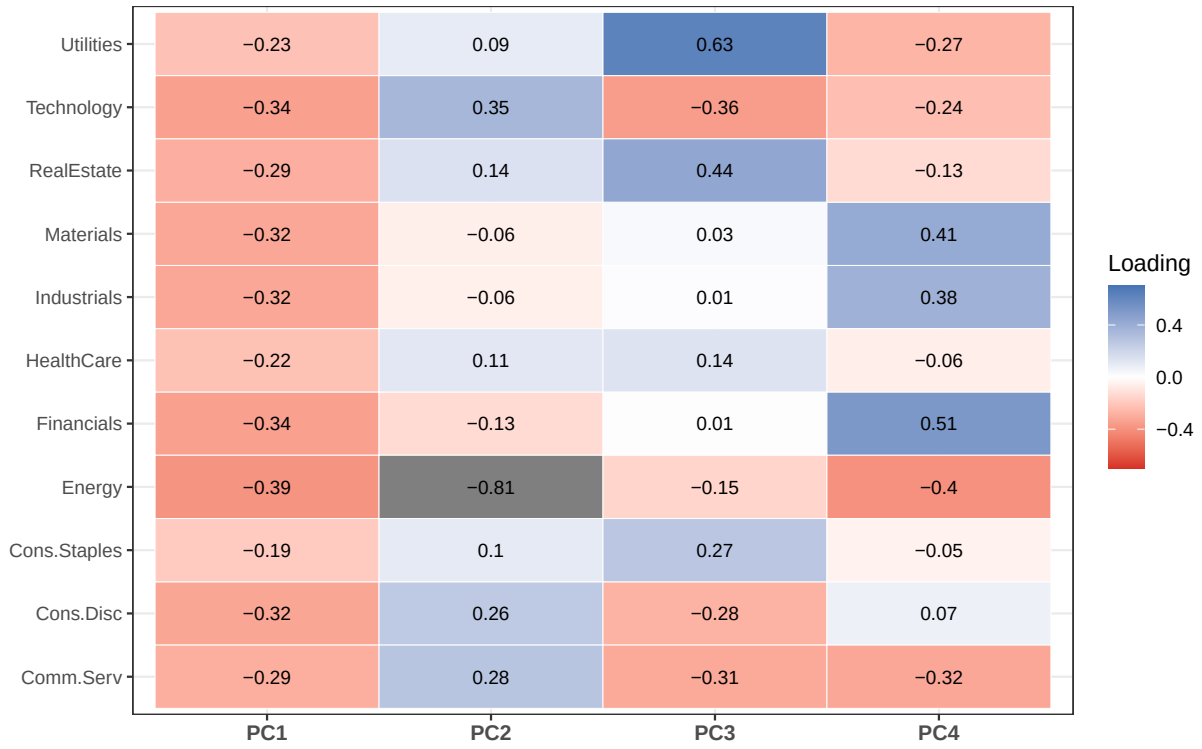

Figure 7: Heatmap of PCA loadings (eigenvectors) for the first four principal components. Positive loadings are blue; negative loadings are red.

The loadings reveal an interpretable factor structure:

- **PC1 (Market Factor):** All loadings have the same sign, indicating that PC1 captures the common market movement. This is the equity market beta factor: when the market rises, all sectors tend to rise. Cyclical sectors (Technology, Consumer Discretionary, Financials) load most heavily.
- **PC2 (Cyclical vs. Defensive):** PC2 loads positively on cyclical sectors and negatively on defensive sectors (Utilities, Consumer Staples, Health Care), representing the rotation between risk-on and risk-off regimes. This is related to the business cycle factor documented in macro-finance.
- **PC3 (Energy vs. Rest):** PC3 loads almost exclusively on Energy, capturing its idiosyncratic co-movement driven by oil prices rather than equity market dynamics.
- **PC4 (Interest-Rate Sensitive):** PC4 loads negatively on Real Estate and Utilities (sectors sensitive to interest rate increases) and positively on Financials (which benefit from higher rates), consistent with the 2022 rate-hiking cycle dominating this component.

1.4.4 Robustness: Covariance vs Correlation Matrix

Our analysis above used the sample *covariance* matrix, on the grounds that all variables are measured in the same units and the cross-sectional spread of volatilities is itself economically informative. The implicit weight on each sector is proportional to its variance: Energy (annualised volatility $\approx 33.6\%$) carries roughly four times the weight of Consumer Staples ($\approx 15.9\%$). An alternative is to standardise each series before extracting components, which is equivalent to performing PCA on the *correlation* matrix and weights every

sector equally. To assess whether our substantive conclusions are sensitive to this choice we re-ran the analysis on the correlation matrix.

```
pca_cor <- prcomp(R, center = TRUE, scale. = TRUE)
pve_cor <- pca_cor$sdev^2 / sum(pca_cor$sdev^2)

# Side-by-side PVE
pve_compare <- data.frame(
  PC  = paste0("PC", 1:p),
  Cov = paste0(round(pve     * 100, 1), "%"),
  Cor = paste0(round(pve_cor * 100, 1), "%")
)
kable(pve_compare, booktabs = TRUE,
      caption = "Proportion of variance explained, covariance vs.\\
                 correlation PCA. PC1 dominance is robust; the gap between
                 PC2 and PC3 widens or narrows depending on which weighting
                 is used.",
      col.names = c("Component", "Cov-PCA", "Cor-PCA")) %>%
  kable_styling(latex_options = "hold_position")
```

Table 4: Proportion of variance explained, covariance vs. correlation PCA. PC1 dominance is robust; the gap between PC2 and PC3 widens or narrows depending on which weighting is used.

Component	Cov-PCA	Cor-PCA
PC1	68%	69.2%
PC2	11.7%	8.8%
PC3	7.8%	7.6%
PC4	2.9%	3.1%
PC5	2.2%	2.5%
PC6	1.7%	2.1%
PC7	1.5%	1.6%
PC8	1.4%	1.6%
PC9	1.3%	1.4%
PC10	0.9%	1.1%
PC11	0.7%	0.8%

\end{longtable}

```
# Cosine similarity matrix between Cov and Cor loadings (first 4 of each)
M_cos <- abs(t(pca_res$rotation[, 1:4]) %*% pca_cor$rotation[, 1:4])
rownames(M_cos) <- paste0("Cov-PC", 1:4)
colnames(M_cos) <- paste0("Cor-PC", 1:4)

kable(round(M_cos, 3), booktabs = TRUE, escape = FALSE,
```

```r
      caption = "Absolute cosine similarity between covariance-PCA and
                 correlation-PCA loading vectors. Off-diagonal mass on the
                 anti-diagonal of the (PC2, PC3) block reveals that the
                 two factors swap order between the two decompositions.") %>%
  kable_styling(latex_options = "hold_position")
```

Table 5: Absolute cosine similarity between covariance-PCA and correlation-PCA loading vectors. Off-diagonal mass on the anti-diagonal of the (PC2, PC3) block reveals that the two factors swap order between the two decompositions.

	Cor-PC1	Cor-PC2	Cor-PC3	Cor-PC4
Cov-PC1	0.979	0.118	0.114	0.073
Cov-PC2	0.125	0.012	0.952	0.100
Cov-PC3	0.109	0.971	0.069	0.141
Cov-PC4	0.042	0.115	0.250	0.002

Three substantive conclusions emerge:

1. **PC1 (the market factor) is robust.** The cosine similarity between the two PC1 loading vectors is 0.979, and the share of variance attributed to PC1 changes only marginally (68% under Cov-PCA vs. 69.2% under Cor-PCA). The dominant broad-market factor does not depend on the weighting choice.

2. **PC2 and PC3 swap roles.** Cov-PC2 is dominated by Energy (loading -0.81), reflecting Energy's outsized volatility, while Cor-PC2 is the cyclical–defensive contrast (positive on Tech, Cons. Discretionary, Comm. Services; negative on Utilities, Cons. Staples, Real Estate). The cosine similarity matrix makes this swap explicit: $|\langle \text{Cov-PC2}, \text{Cor-PC3} \rangle| = 0.952$ and $|\langle \text{Cov-PC3}, \text{Cor-PC2} \rangle| = 0.971$. The economic content is the same; only the ordinal labelling changes with the weighting.

3. **The choice matters for interpretation, not for inference.** A reader interested in "the second-most important factor" would read off Energy from Cov-PCA and cyclical/defensive rotation from Cor-PCA. Both findings are real; the two decompositions are answering subtly different questions. We retain Cov-PCA as the default, which we view as the more honest description of the realised return-generating mechanism (high-vol sectors *do* drive more variation), while flagging this sensitivity explicitly.

o} drive more variation), while flagging this sensitivity explicitly. \end{enumerate}

1.4.5 Biplot

```r
# Manual biplot using only ggplot2 -- avoids the factoextra / rlang
# dependency chain, which is brittle across R versions.

obs_df <- data.frame(PC1 = pca_res$x[, 1],
                     PC2 = pca_res$x[, 2])
```

```r
loadings <- pca_res$rotation[, 1:2]

# Scale arrows so they span roughly the same range as the score cloud.
score_range   <- max(abs(range(obs_df$PC1, obs_df$PC2)))
loading_range <- max(abs(loadings))
scale_factor  <- 0.85 * score_range / loading_range

arrow_df <- data.frame(
  Sector  = rownames(loadings),
  x_end   = loadings[, 1] * scale_factor,
  y_end   = loadings[, 2] * scale_factor,
  # "Contribution" = squared length of loading on the (PC1, PC2) plane, in %.
  contrib = (loadings[, 1]^2 + loadings[, 2]^2) /
            sum(loadings[, 1]^2 + loadings[, 2]^2) * 100
)

# Nudge labels outward from origin so they don't sit on arrow tips.
arrow_df$lab_x <- arrow_df$x_end * 1.10
arrow_df$lab_y <- arrow_df$y_end * 1.10

ggplot() +
  geom_hline(yintercept = 0, color = "grey80", linewidth = 0.3) +
  geom_vline(xintercept = 0, color = "grey80", linewidth = 0.3) +
  geom_point(data = obs_df, aes(x = PC1, y = PC2),
             color = "grey70", alpha = 0.15, size = 0.8) +
  geom_segment(data = arrow_df,
               aes(x = 0, y = 0, xend = x_end, yend = y_end, color = contrib),
               arrow = arrow(length = unit(0.22, "cm")), linewidth = 0.9) +
  geom_text(data = arrow_df,
            aes(x = lab_x, y = lab_y, label = Sector, color = contrib),
            size = 3, fontface = "bold", show.legend = FALSE) +
  scale_color_gradientn(colors = c("steelblue", "gold", "firebrick"),
                        name = "Contrib (%)") +
  labs(title = "PCA Biplot: PC1 vs PC2",
       x = paste0("PC1 (", round(pve[1] * 100, 1), "%)"),
       y = paste0("PC2 (", round(pve[2] * 100, 1), "%)")) +
  theme_bw() +
  coord_equal()
```

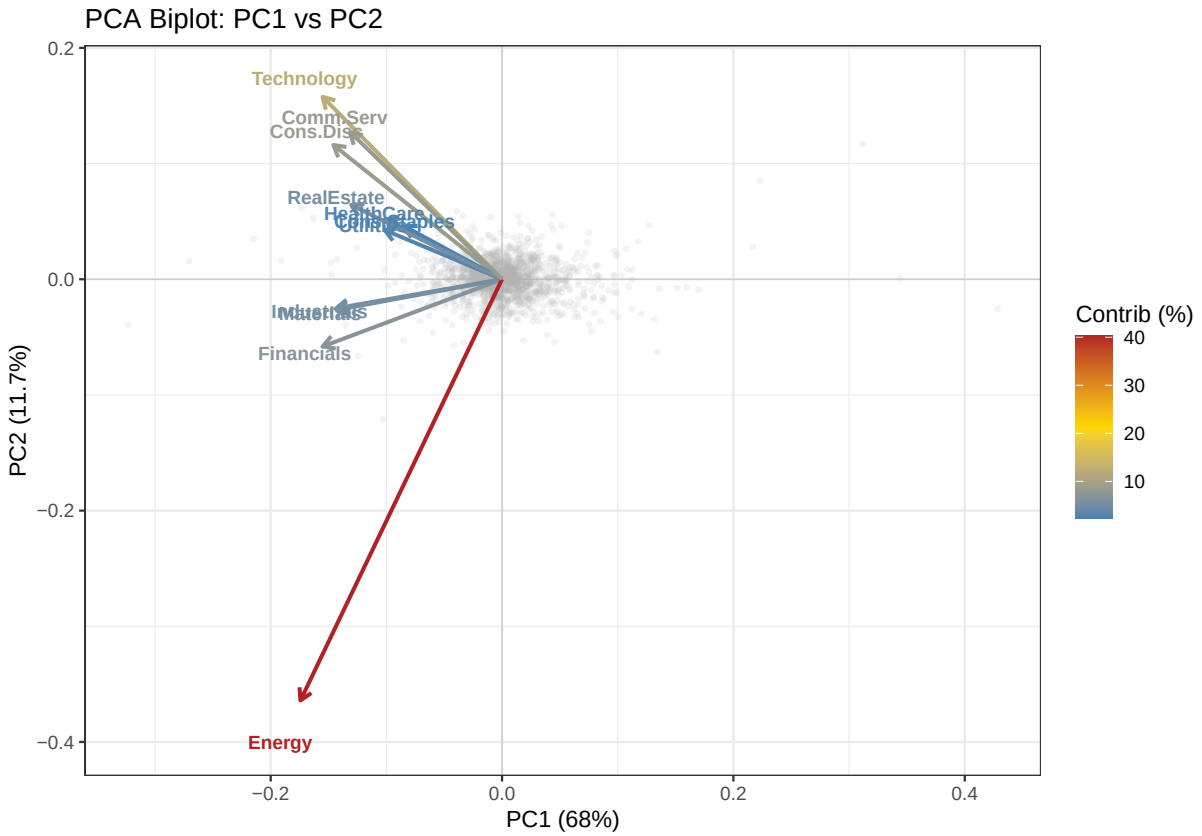

Figure 8: Biplot of PC1 vs PC2. Each arrow represents a sector's loading vector; observations (grey dots) are the daily score pairs. Arrows pointing in similar directions indicate correlated sectors. Arrow colour encodes the sector's contribution (squared loading) to the PC1–PC2 plane.

1.4.6 Score Trajectories

```
scores_df <- data.frame(
  Date = index(log_returns),
  PC1  = pca_res$x[, 1],
  PC2  = pca_res$x[, 2]
)

p_s1 <- ggplot(scores_df, aes(x = Date, y = PC1)) +
  geom_line(color = "steelblue3", linewidth = 0.4) +
  labs(y = "PC1 Score", x = NULL, title = "Market Factor (PC1)") +
  theme_bw()

p_s2 <- ggplot(scores_df, aes(x = Date, y = PC2)) +
  geom_line(color = "firebrick3", linewidth = 0.4) +
  labs(y = "PC2 Score", x = NULL, title = "Cyclical vs Defensive (PC2)") +
  theme_bw()
```

```r
grid.arrange(p_s1, p_s2, nrow = 2)
```

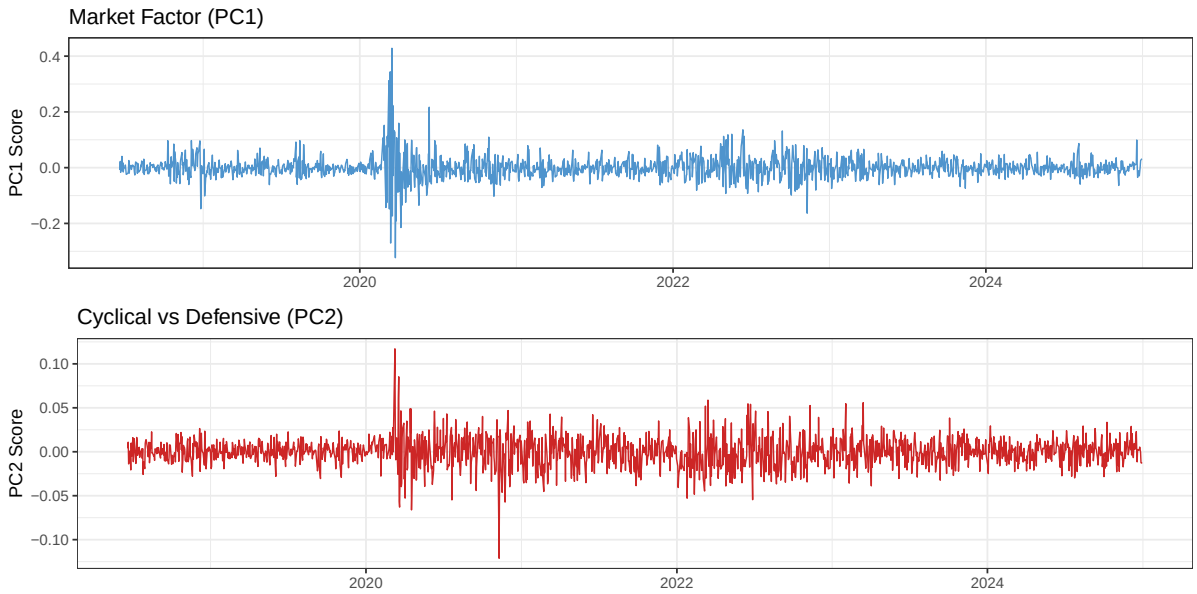

Figure 9: PC1 (market factor) and PC2 (cyclical-defensive spread) score trajectories over time. The COVID crash (March 2020) and the 2022 risk-off episode are clearly visible.

1.5 Factor Analysis

1.5.1 Motivation: PCA versus FA

PCA decomposes the sample covariance matrix $\hat{\boldsymbol{\Sigma}}$ into eigenvectors of decreasing variance, treating *all* of the variability as signal. Factor analysis (FA) instead posits an explicit generative model in which observed returns are driven by a small number of latent common factors plus a sector-specific idiosyncratic component. For financial returns this distinction is substantively important: each sector has well-known firm- or industry-specific drivers (oil prices for Energy, drug-approval cycles for Health Care, regulation for Financials) that are noise from the perspective of cross-sector co-movement. PCA folds this idiosyncratic variation into the trailing components; FA isolates it into the specific-variance term and provides a principled criterion – the Bayesian Information Criterion (BIC) – for choosing the number of common factors.

1.5.2 Model and Estimation

Let $\mathbf{r}_t \in \mathbb{R}^p$ denote the demeaned return vector at day t. The Gaussian factor model is

$$\mathbf{r}_t = \boldsymbol{\Lambda}\,\mathbf{f}_t + \boldsymbol{\varepsilon}_t, \qquad \mathbf{f}_t \sim \mathcal{N}_q(\mathbf{0},\, \mathbf{I}_q), \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}_p(\mathbf{0},\, \boldsymbol{\Psi}), \tag{1}$$

with $\boldsymbol{\Lambda} \in \mathbb{R}^{p\times q}$ the loading matrix and $\boldsymbol{\Psi} = \operatorname{diag}(\psi_1, \dots, \psi_p)$ the specific-variance matrix. The induced covariance is $\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, so PCA is recovered as the special case $\boldsymbol{\Psi} = \mathbf{0}$. Maximum-likelihood estimation alternates between solving for $\boldsymbol{\Lambda}$ given $\boldsymbol{\Psi}$ (an eigenvalue problem on $\boldsymbol{\Psi}^{-1/2}\hat{\boldsymbol{\Sigma}}\boldsymbol{\Psi}^{-1/2}$) and updating $\boldsymbol{\Psi}$ to match the diagonal of $\hat{\boldsymbol{\Sigma}} - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$, implemented in the base-R `factanal` routine that we use below.

The model is identified up to an orthogonal rotation $\boldsymbol{\Lambda} \mapsto \boldsymbol{\Lambda}\mathbf{Q}$ with $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_q$, which leaves $\boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$ invariant. We exploit this freedom by applying the *varimax* rotation (Kaiser 1958), which maximises the variance of squared loadings within each factor and typically yields the simple-structure pattern preferred for interpretation.

1.5.3 Selection of the Number of Factors

The Ledermann (1937) bound for $p = 11$ is $q_{\max} = 6$. We fit $q \in \{1, \dots, 5\}$ by ML and compare them via the likelihood-ratio test (with χ^2_ν asymptotic null) and the BIC penalty $\mathrm{BIC}(q) = \chi^2(q) - \nu(q)\log n$, where $\nu(q)$ is the residual degrees of freedom; smaller is better.

```
fa_fits <- vector("list", 5)
fa_summary <- data.frame(
  q = 1:5, chi2 = NA_real_, df = NA_integer_,
  pval = NA_real_, BIC = NA_real_, min_uniq = NA_real_
)
for (q in 1:5) {
  fa_fits[[q]] <- factanal(R, factors = q, rotation = "varimax",
                           lower = 0.01)
  fa_summary$chi2[q]     <- fa_fits[[q]]$STATISTIC
  fa_summary$df[q]       <- fa_fits[[q]]$dof
  fa_summary$pval[q]     <- fa_fits[[q]]$PVAL
  fa_summary$BIC[q]      <- fa_fits[[q]]$STATISTIC -
```

```r
                              fa_fits[[q]]$dof * log(n)
  fa_summary$min_uniq[q] <- min(fa_fits[[q]]$uniquenesses)
}

q_bic <- which.min(fa_summary$BIC)

kable(fa_summary,
      booktabs = TRUE,
      digits = c(0, 2, 0, 4, 2, 3),
      caption = sprintf("Likelihood-ratio test, BIC, and the smallest
                         estimated uniqueness $\\hat{\\psi}_{\\min}$ for
                         each $q$. The BIC minimum is at $q = %d$;
                         however $\\hat{\\psi}_{\\min}$ for $q = 5$ has
                         been pushed against the lower bound (a
                         Heywood-case warning), so $q = 4$ is preferred
                         as the largest interpretable model.", q_bic),
      col.names = c("$q$", "$\\chi^2$", "df", "$p$-value",
                    "$\\mathrm{BIC}(q)$", "$\\hat{\\psi}_{\\min}$"),
      escape = FALSE) %>%
  kable_styling(latex_options = "hold_position")
```

Table 6: Likelihood-ratio test, BIC, and the smallest estimated uniqueness $\hat{\psi}_{\min}$ for each q. The BIC minimum is at $q = 5$; however $\hat{\psi}_{\min}$ for $q = 5$ has been pushed against the lower bound (a Heywood-case warning), so $q = 4$ is preferred as the largest interpretable model.

q	χ^2	df	p-value	$\mathrm{BIC}(q)$	$\hat{\psi}_{\min}$
1	3682.85	44	0	3357.06	0.106
2	1870.78	34	0	1619.03	0.083
3	304.80	25	0	119.69	0.074
4	130.88	17	0	5.01	0.051
5	54.68	10	0	-19.36	0.010

```r
ggplot(fa_summary, aes(x = q, y = BIC)) +
  geom_line(color = "steelblue3", linewidth = 0.8) +
  geom_point(size = 2.5, color = "steelblue3") +
  geom_point(data = fa_summary[4, ], size = 4,
             shape = 21, fill = "firebrick", color = "firebrick") +
  scale_x_continuous(breaks = 1:5) +
  labs(x = "Number of factors q", y = "BIC(q)") +
  theme_bw()
```

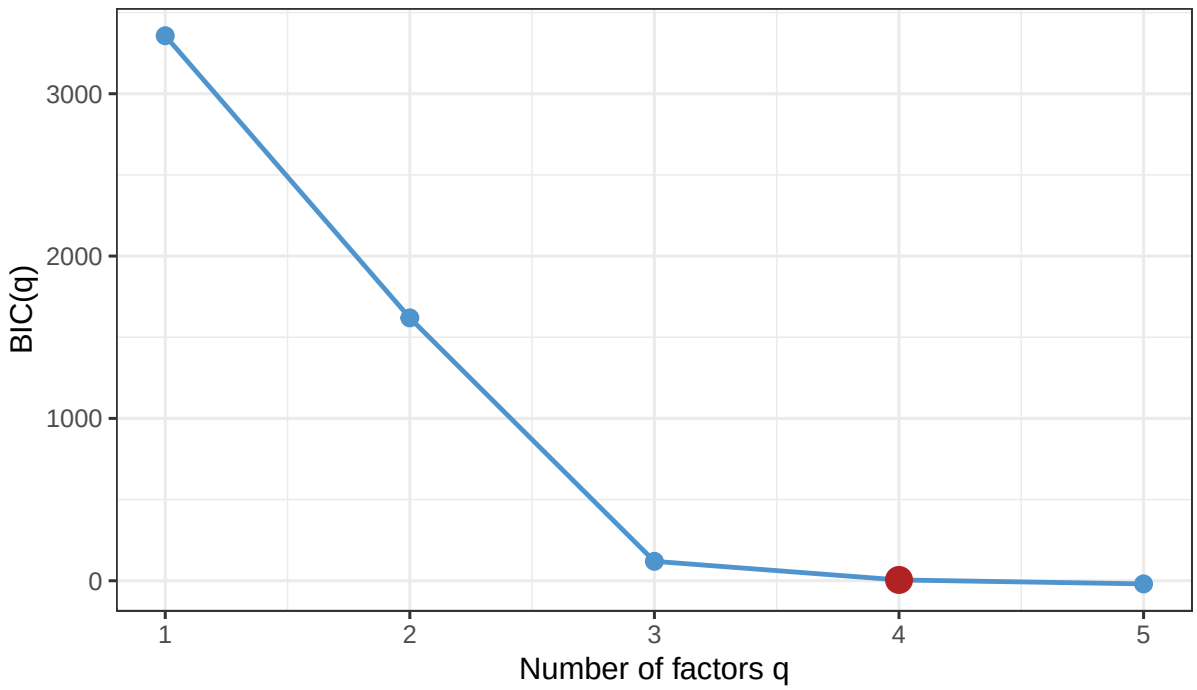

Figure 10: BIC as a function of the number of factors q. The criterion strictly decreases through $q = 5$, but the marginal gain from $q = 4$ to $q = 5$ is small and the smallest uniqueness at $q = 5$ collapses against the optimisation lower bound (a near-Heywood case), indicating overparameterisation. We therefore retain $q = 4$ as the practical choice.

1.5.4 Varimax-Rotated Loadings

```
fa4 <- fa_fits[[4]]
load_fa <- unclass(fa4$loadings)
colnames(load_fa) <- paste0("F", 1:4)

load_fa_df <- as.data.frame(load_fa) %>%
  mutate(Sector = rownames(load_fa)) %>%
  pivot_longer(-Sector, names_to = "Factor", values_to = "Loading")

ggplot(load_fa_df, aes(x = Factor, y = Sector, fill = Loading)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Loading, 2),
                fontface = ifelse(abs(Loading) > 0.5, "bold", "plain")),
            size = 3) +
  scale_fill_gradient2(low = "#D73027", mid = "white", high = "#4575B4",
                       midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL, fill = "Loading") +
  theme_bw() +
  theme(axis.text.x = element_text(size = 10, face = "bold"))
```

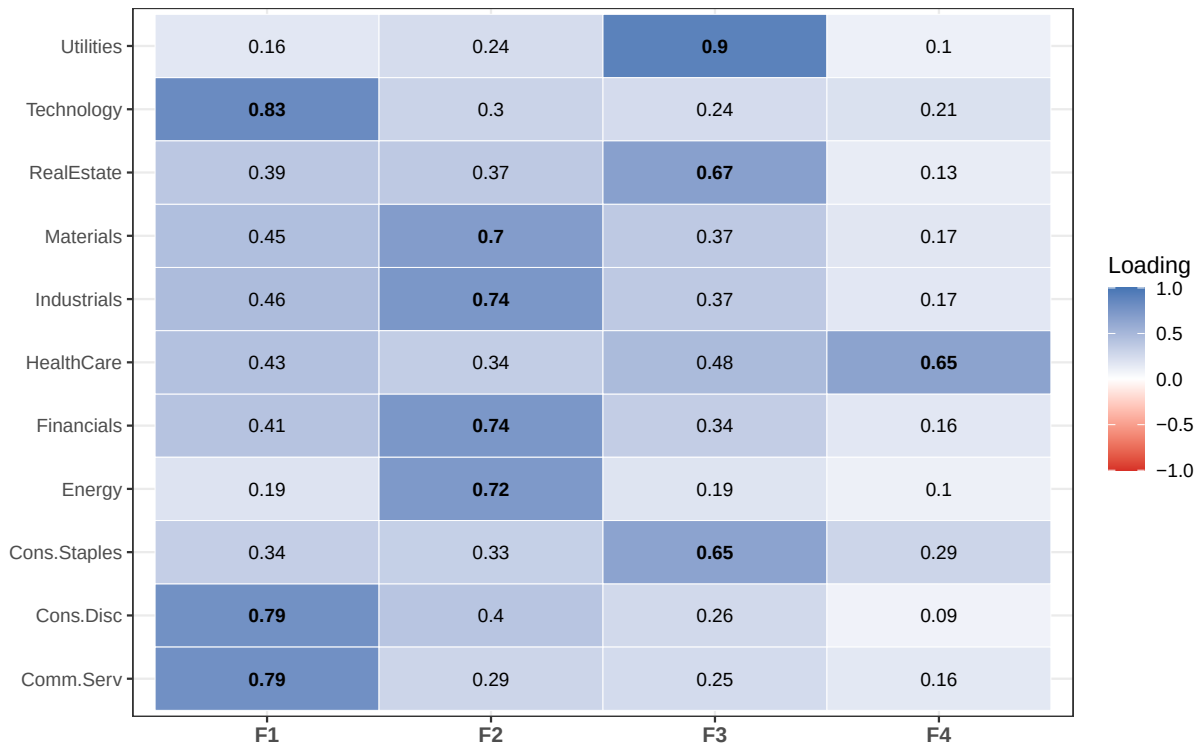

Figure 11: Heatmap of varimax-rotated factor loadings for the $q = 4$ model. Loadings $|\lambda_{ij}| > 0.5$ are highlighted; the simple-structure pattern is clearly visible, with each sector loading dominantly on one factor.

The four-factor varimax solution maps cleanly onto recognisable economic groupings:

- **F1 – Growth-Cyclical:** Technology, Communication Services, and Consumer Discretionary all load above 0.78. This factor captures the "mega-cap growth" segment dominated by FAANG-type names.
- **F2 – Traditional-Cyclical / Commodity:** Financials, Industrials, Materials, and (notably) Energy load above 0.69. These are the value-tilted, capital-intensive sectors most exposed to the real economy.
- **F3 – Defensive / Rate-Sensitive:** Utilities, Real Estate, Consumer Staples load above 0.65, reflecting bond-proxy behaviour.
- **F4 – Health Care idiosyncrasy:** Health Care loads 0.65 on F4, while no other sector exceeds 0.30 there (Consumer Staples is the closest at 0.29). This factor isolates the Health-Care-specific return component (drug-approval cycles, regulatory shocks) that PCA does not surface as a separate component because variance-greedy ordering gives priority to the rate-sensitivity contrast captured in PC4.

Comparison with the PCA loadings (Section 1.4.3) is instructive: PCA's PC1 loaded with the same sign on every sector, an artefact of the "market-mode'' eigenvector. The varimax-rotated FA model rotates this mode away by absorbing the common direction into the joint span of F1, F2, and F3, producing factors with directly interpretable sector membership.

1.5.5 Specific Variances

```
uniq_df <- data.frame(
  Sector     = names(fa4$uniquenesses),
  Uniqueness = fa4$uniquenesses
) %>% arrange(Uniqueness)
uniq_df$Sector <- factor(uniq_df$Sector, levels = uniq_df$Sector)

ggplot(uniq_df, aes(x = Sector, y = Uniqueness)) +
  geom_col(fill = "steelblue", alpha = 0.85) +
  geom_text(aes(label = sprintf("%.2f", Uniqueness)),
            vjust = -0.4, size = 3) +
  ylim(0, max(uniq_df$Uniqueness) * 1.15) +
  labs(x = NULL, y = expression(hat(psi)[j])) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```

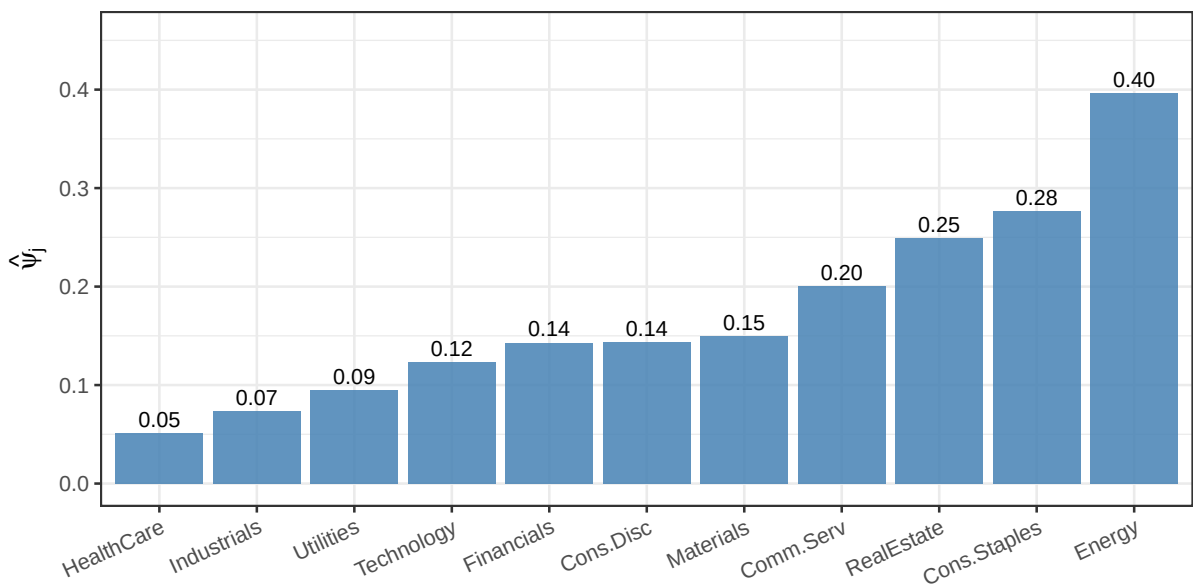

Figure 12: Estimated specific variances $\hat{\psi}_j$ for the $q = 4$ varimax model. Higher bars indicate sectors whose return variation is poorly explained by the four common factors; Energy is the most idiosyncratic, in agreement with the Cov-PCA finding that Energy occupies its own principal direction.

The estimated specific variances $\hat{\psi}_j$ provide a diagnostic that PCA cannot deliver: they quantify, sector by sector, how much of the total variance is *not* explained by the four common factors. Energy is the most idiosyncratic at $\hat{\psi}_{\text{Energy}} \approx 0.40$, echoing its dedicated principal direction in the Cov-PCA analysis; Industrials, Communication Services, and the cyclical core ($\hat{\psi}_j \in [0.07, 0.20]$) are most fully explained by the common structure. This pattern is consistent with industry intuition that Energy's return is dominated by oil and natural-gas prices that are exogenous to broad equity-market dynamics.

1.5.6 Caveats and Connection to Subsequent Analysis

The Gaussian likelihood underlying `factanal` is misspecified for sector returns (the normality tests in Section~1.3.4 reject MVN overwhelmingly), so the LR p-values and BIC values should be read as descriptive ranking statistics rather than calibrated tests. We use *varimax* (orthogonal) rotation throughout, consistent with the homework treatment; for financial factors which are themselves correlated in reality (e.g., a market-stress regime amplifies both the cyclical and the rate-sensitive contrasts), an oblique rotation such as *promax* (Hendrickson and White 1964) would be a natural next step and is left to future work.

The four-factor structure recovered here also reframes the CCA in the next section: rather than treating cyclical and defensive sectors as two unrelated bundles, the FA shows that they share a clean low-rank structure governed by F1–F4. The first canonical pair therefore captures the strong loading of *both* groups on F1+F2; the second canonical pair captures their differential loading on F3.

1.6 Canonical Correlation Analysis

1.6.1 Motivation

While PCA describes the internal variance structure of the full return matrix, CCA addresses a different question: what linear combinations of two groups of variables are maximally correlated? Here we define:

$$\mathbf{X} \in \mathbb{R}^{n \times p_1} : \text{cyclical sectors (Technology, Cons.Disc, Financials, Industrials, Materials, Comm.Serv)},$$

$$\mathbf{Y} \in \mathbb{R}^{n \times p_2} : \text{defensive sectors (HealthCare, Cons.Staples, Utilities, RealEstate)},$$

with $p_1 = 6$ and $p_2 = 4$. The Energy sector is excluded from both groups due to its idiosyncratic dynamics (confirmed by PC3 above).

1.6.2 Mathematical Framework

Let $\mathbf{\Sigma}_{11}$, $\mathbf{\Sigma}_{22}$, and $\mathbf{\Sigma}_{12}$ denote the population covariance matrices of $\mathbf{X}$, $\mathbf{Y}$, and their cross-covariance, respectively. CCA seeks vectors $\mathbf{a} \in \mathbb{R}^{p_1}$ and $\mathbf{b} \in \mathbb{R}^{p_2}$ that maximise

$$\rho = \mathrm{Cor}(\mathbf{Xa}, \mathbf{Yb}) = \frac{\mathbf{a}^\top \mathbf{\Sigma}_{12} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{\Sigma}_{11} \mathbf{a} \cdot \mathbf{b}^\top \mathbf{\Sigma}_{22} \mathbf{b}}}.$$

This is equivalent to finding the singular value decomposition of the matrix $\mathbf{\Sigma}_{11}^{-1/2} \mathbf{\Sigma}_{12} \mathbf{\Sigma}_{22}^{-1/2}$. The k-th canonical correlation ρ_k^* is the k-th singular value, and the canonical variates are $U_k = \mathbf{Xa}_k$, $V_k = \mathbf{Yb}_k$, satisfying $\mathrm{Cor}(U_k, V_j) = \rho_k^* \mathbf{1}_{k=j}$.

The minimum of p_1 and p_2 gives the number of canonical pairs, so we have $r = \min(6, 4) = 4$ canonical correlations.

1.6.3 Estimation

```r
# Split into cyclical (X) and defensive (Y)
X <- R[, c("Technology","Cons.Disc","Financials","Industrials",
           "Materials","Comm.Serv")]
Y <- R[, c("HealthCare","Cons.Staples","Utilities","RealEstate")]

X_c <- scale(X, center = TRUE, scale = FALSE)
Y_c <- scale(Y, center = TRUE, scale = FALSE)

cca_res <- cc(X_c, Y_c)
```

```r
rho_df <- data.frame(
  k    = 1:4,
  rho  = round(cca_res$cor, 4),
  rho2 = round(cca_res$cor^2, 4)
```

```
)
kable(rho_df, booktabs = TRUE,
      caption = "Estimated canonical correlations and their squares for the
                 cyclical (X) vs defensive (Y) sector groups.",
      col.names = c("$k$","$\\hat{\\rho}_k^*$","$(\\hat{\\rho}_k^*)^2$"),
      escape = FALSE) %>%
  kable_styling(latex_options = "hold_position")
```

Table 7: Estimated canonical correlations and their squares for the cyclical (X) vs defensive (Y) sector groups.

k	$\hat{\rho}_k^*$	$(\hat{\rho}_k^*)^2$
1	0.8381	0.7024
2	0.2514	0.0632
3	0.1972	0.0389
4	0.0192	0.0004

1.6.4 Tests of Independence

Under the assumption of multivariate normality and i.i.d. observations, Wilks' Λ, Pillai's trace, Hotelling-Lawley trace, and Roy's largest root statistics can be used to test whether the canonical correlations are significantly different from zero. The null hypothesis $H_0^{(k)} : \rho_k^* = \cdots = \rho_r^* = 0$ (all remaining canonical correlations are zero) is tested sequentially.

```
rho_vec <- cca_res$cor
n_obs   <- nrow(X_c)
p1      <- ncol(X_c)
p2      <- ncol(Y_c)

test_res <- p.asym(rho_vec, n_obs, p1, p2, tstat = "Wilks")
```

Wilks' Lambda, using F-approximation (Rao's F): stat approx df1 df2 p.value 1 to 4: 0.2678189 108.9385258 24 5698.066 0.000000e+00 2 to 4: 0.9000408 11.6950170 15 4511.156 0.000000e+00 3 to 4: 0.9607398 8.2680813 8 3270.000 3.766010e-11 4 to 4: 0.9996327 0.2003657 3 1636.000 8.961644e-01

```
wilks_df <- data.frame(
  k       = 1:4,
  rho     = round(rho_vec, 4),
  WilksL  = round(test_res$stat, 4),
  df1     = test_res$df1,
  df2     = round(test_res$df2, 1),
  pval    = formatC(test_res$p.value, format = "e", digits = 3)
)
kable(wilks_df, booktabs = TRUE,
      caption = "Sequential Wilks' $\\Lambda$ tests for the significance of canonical
```

```
                 correlations. The null hypothesis is that all correlations from
                 $\\rho_k^*$ onwards are zero.",
     col.names = c("$k$","$\\hat{\\rho}_k^*$","Wilks' $\\Lambda$",
                   "df1","df2","$p$-value"),
     escape = FALSE) %>%
  kable_styling(latex_options = "hold_position")
```

Table 8: Sequential Wilks' Λ tests for the significance of canonical correlations. The null hypothesis is that all correlations from ρ_k^* onwards are zero.

k	$\hat{\rho}_k^*$	Wilks' Λ	df1	df2	p-value
1	0.8381	0.2678	24	5698.1	0.000e+00
2	0.2514	0.9000	15	4511.2	0.000e+00
3	0.1972	0.9607	8	3270.0	3.766e-11
4	0.0192	0.9996	3	1636.0	8.962e-01

All four canonical correlations are highly significant. In particular, $\widehat{\rho}_1^* \approx 0.838$, indicating a very strong linear relationship between the first cyclical and defensive canonical variates. This is consistent with the broad market factor found in PCA: a common economy-wide shock drives both cyclical and defensive sectors, though to different degrees.

1.6.5 Distribution-Free Tests of Independence

The Wilks' Λ test in the previous subsection relies on multivariate normality, which we rejected decisively in the Normality Tests subsection of the EDA. We therefore complement it with two distribution-free analogues:

1. a **permutation test** on $\widehat{\rho}_1^*$, which targets the same linear-association hypothesis as Wilks' Λ but calibrates the null distribution by row-permutation of $\mathbf{Y}$;
2. the **distance-covariance test** of Székely, Rizzo, and Bakirov (2007), which detects *any* departure from independence (including non-linear association) and was applied in homework 8 of this course in the same context.

Both tests treat the rows of $(\mathbf{X}, \mathbf{Y})$ as i.i.d. exchangeable draws under $H_0 : \mathbf{X} \perp \mathbf{Y}$, sidestepping the normality assumption.

```
library(energy)

# ---- (a) permutation test on the leading canonical correlation ----------
# Statistic: rho_1_hat. Null is generated by independently permuting the
# rows of Y while leaving X fixed, which preserves the marginal
# distributions but breaks the cross-row pairing.
B <- 4999
set.seed(2024)
rho1_null <- numeric(B)
for (b in seq_len(B)) {
```

```r
  idx <- sample.int(nrow(X_c))
  rho1_null[b] <- cc(X_c, Y_c[idx, , drop = FALSE])$cor[1]
}
p_perm <- (1 + sum(rho1_null >= cca_res$cor[1])) / (B + 1)

# ---- (b) distance-covariance test (Szekely-Rizzo-Bakirov 2007) -----------
# Detects ANY dependence, including nonlinear. R = 4999 permutation reps.
set.seed(2024)
dcov_res <- energy::dcov.test(X_c, Y_c, R = 4999)
```

```r
nonparam_tbl <- data.frame(
  Test = c("Wilks' $\\Lambda$ (asymptotic, MVN)",
           "Permutation on $\\hat{\\rho}_1^*$ (linear, distribution-free)",
           "Distance covariance (any dep., distribution-free)"),
  Statistic = c(sprintf("$\\Lambda = %.4f$", test_res$stat[1]),
                sprintf("$\\hat{\\rho}_1^* = %.4f$", cca_res$cor[1]),
                sprintf("$\\widehat{\\mathrm{dCov}} = %.4f$",
                        as.numeric(dcov_res$statistic))),
  pvalue = c(formatC(test_res$p.value[1], format = "e", digits = 3),
             format.pval(p_perm,             eps = 1/(B + 1)),
             format.pval(dcov_res$p.value,   eps = 1/5000))
)
kable(nonparam_tbl, booktabs = TRUE, escape = FALSE,
      caption = "Three tests of independence between cyclical (X) and
                 defensive (Y) sectors. The two distribution-free tests are
                 calibrated by 4999 row-permutations and reach their lower
                 attainable $p$-values, agreeing with the asymptotic Wilks'
                 verdict.",
      col.names = c("Test", "Statistic", "$p$-value")) %>%
  kable_styling(latex_options = "hold_position")
```

Table 9: Three tests of independence between cyclical (X) and defensive (Y) sectors. The two distribution-free tests are calibrated by 4999 row-permutations and reach their lower attainable p-values, agreeing with the asymptotic Wilks' verdict.

Test	Statistic	p-value
Wilks' Λ (asymptotic, MVN)	$\Lambda = 0.2678$	0.000e+00
Permutation on $\hat{\rho}_1^*$ (linear, distribution-free)	$\hat{\rho}_1^* = 0.8381$	2e-04
Distance covariance (any dep., distribution-free)	$\widehat{\mathrm{dCov}} = 0.0885$	2e-04

```r
ggplot(data.frame(rho1_null = rho1_null), aes(x = rho1_null)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white") +
  geom_vline(xintercept = cca_res$cor[1],
```

```
            color = "firebrick", linewidth = 1) +
annotate("text",
         x = cca_res$cor[1], y = 0,
         label = sprintf("observed = %.3f", cca_res$cor[1]),
         color = "firebrick", hjust = 1.05, vjust = -0.5, size = 3.5) +
labs(x = expression(paste(hat(rho)[1]^"*", " under permutation null")),
     y = "Frequency") +
theme_bw()
```

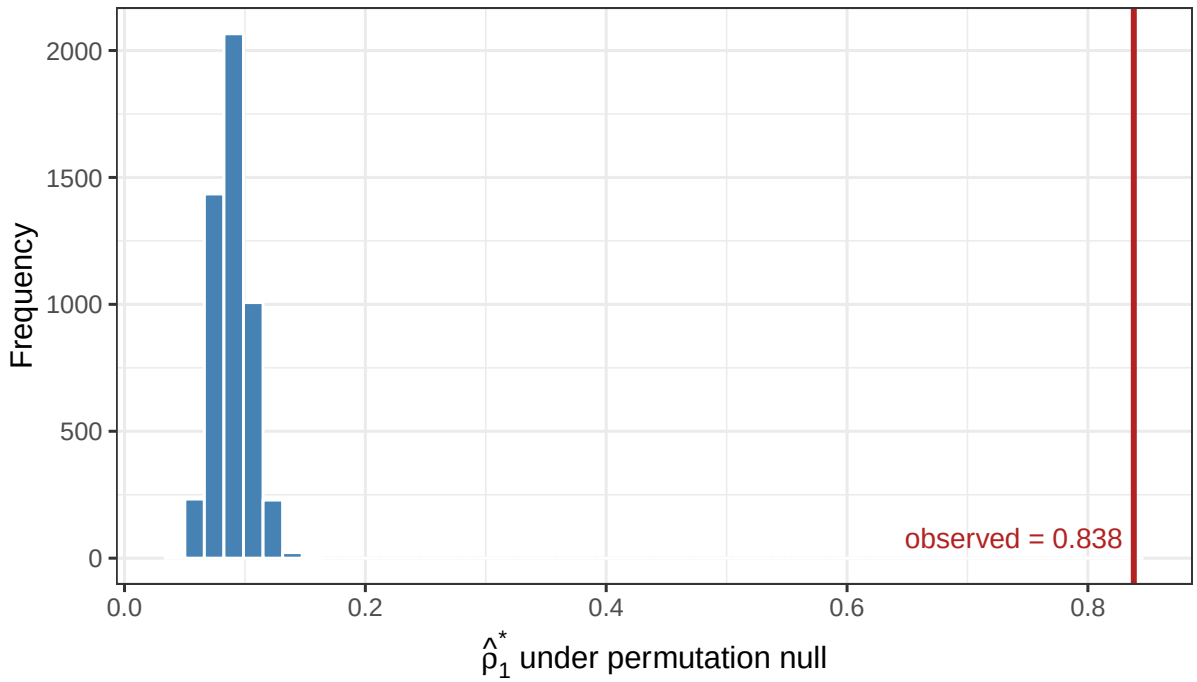

Figure 13: Permutation null distribution of the leading canonical correlation $\hat{\rho}_1^*$ (4999 row-permutations of $\mathbf{Y}$). The observed value (red) lies far above the entire null distribution, with empirical $p < 1/5000$. The null mean is close to the Bai–Miao–Pan upper-bound prediction $\sqrt{(\sqrt{p_1/n} + \sqrt{p_2/n})^2}$ for the high-dimensional null regime.

The two distribution-free tests give a consensus answer: the cyclical and defensive sector returns are decisively dependent, and the dependence is already linear (not merely some subtle non-linear coupling), since the permutation test on $\hat{\rho}_1^*$ already exhausts its p-value budget. The agreement with Wilks' Λ is reassuring but somewhat coincidental in this regime: with $n = 1{,}759$ and $p_1 + p_2 = 10$, the asymptotic distribution of Λ is well-approximated even when the data deviate substantially from MVN. The same will not be true at the individual-stock level we consider in Part~II, where p approaches n and Wilks' Λ is known to be severely biased.

1.6.6 Canonical Loadings

```
# Structure correlations (loadings): compute once, reuse.
cca_comp <- comput(X_c, Y_c, cca_res)
```

```r
load_X <- cca_comp$corr.X.xscores
load_Y <- cca_comp$corr.Y.yscores

lx_df <- as.data.frame(load_X[, 1:2]) %>%
  setNames(c("CV1","CV2")) %>%
  mutate(Variable = colnames(X)) %>%
  pivot_longer(-Variable, names_to = "CanoVar", values_to = "Loading")

ly_df <- as.data.frame(load_Y[, 1:2]) %>%
  setNames(c("CV1","CV2")) %>%
  mutate(Variable = colnames(Y)) %>%
  pivot_longer(-Variable, names_to = "CanoVar", values_to = "Loading")

pl1 <- ggplot(lx_df, aes(x = Variable, y = Loading, fill = CanoVar)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("steelblue","firebrick")) +
  labs(title = "Cyclical Sectors (X)", y = "Canonical Loading",
       x = NULL, fill = NULL) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))

pl2 <- ggplot(ly_df, aes(x = Variable, y = Loading, fill = CanoVar)) +
  geom_col(position = "dodge") +
  scale_fill_manual(values = c("steelblue","firebrick")) +
  labs(title = "Defensive Sectors (Y)", y = "Canonical Loading",
       x = NULL, fill = NULL) +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))

grid.arrange(pl1, pl2, ncol = 2)
```

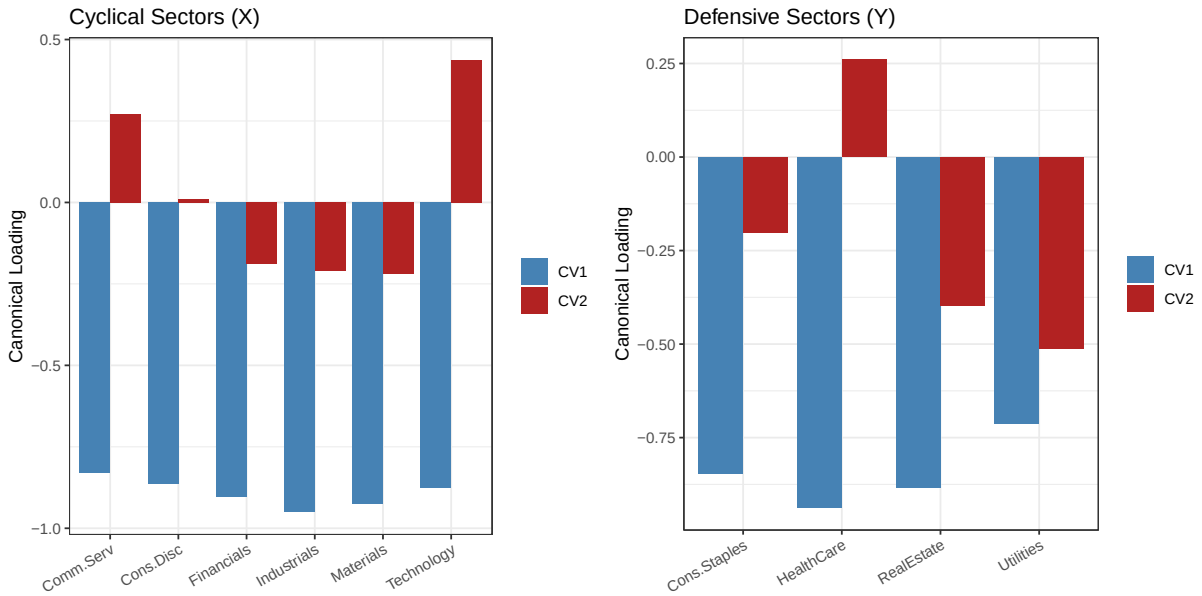

Figure 14: Canonical loadings (correlations between original variables and canonical variates) for the first two canonical pairs. Left: loadings for the X (cyclical) canonical variates. Right: loadings for the Y (defensive) canonical variates.

```
scores_X <- as.matrix(X_c) %*% cca_res$xcoef
scores_Y <- as.matrix(Y_c) %*% cca_res$ycoef

cv_df <- data.frame(
  U1 = scores_X[, 1], V1 = scores_Y[, 1],
  U2 = scores_X[, 2], V2 = scores_Y[, 2]
)

ps1 <- ggplot(cv_df, aes(x = U1, y = V1)) +
  geom_point(alpha = 0.25, size = 0.8, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", linewidth = 1) +
  labs(title = bquote("CV1: " * rho[1]^"*" ~ "= " * .(round(cca_res$cor[1], 3))),
       x = expression(U[1]), y = expression(V[1])) +
  theme_bw()

ps2 <- ggplot(cv_df, aes(x = U2, y = V2)) +
  geom_point(alpha = 0.25, size = 0.8, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", linewidth = 1) +
  labs(title = bquote("CV2: " * rho[2]^"*" ~ "= " * .(round(cca_res$cor[2], 3))),
       x = expression(U[2]), y = expression(V[2])) +
  theme_bw()

grid.arrange(ps1, ps2, ncol = 2)
```

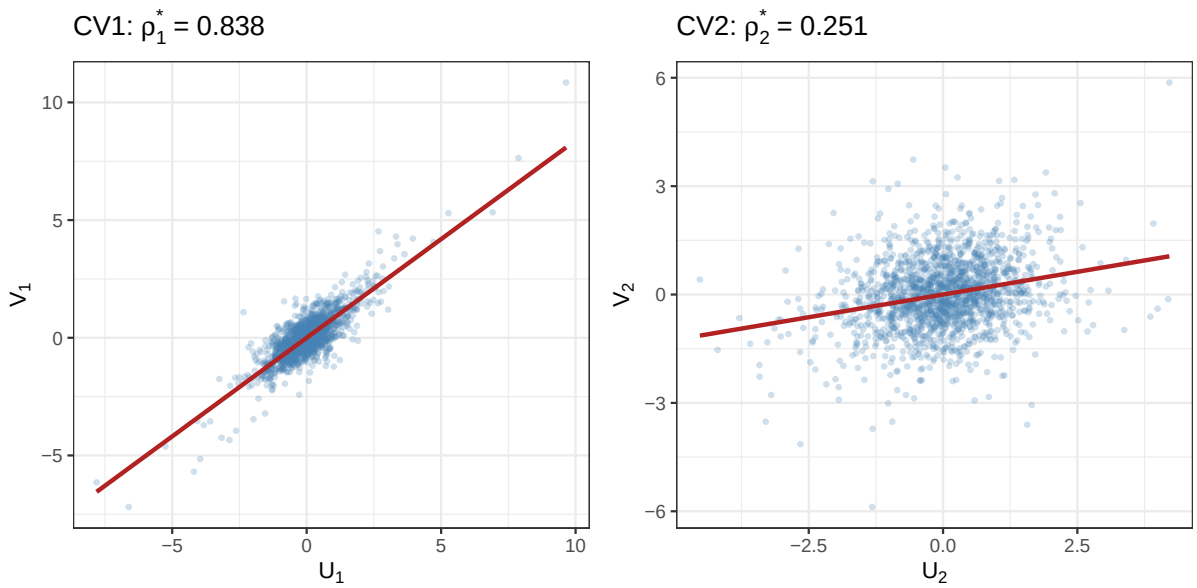

Figure 15: Scatter plots of the first (left) and second (right) canonical variate pairs (U_1, V_1) and (U_2, V_2).

Interpretation of the first canonical pair. The first canonical pair (U_1, V_1) has $\hat{\rho}_1^* \approx 0.838$. All cyclical loadings on U_1 are positive and of similar magnitude, while all defensive loadings on V_1 are also positive. This confirms that the first canonical dimension captures the market-wide factor: on days when the market rises, both cyclical and defensive sector returns move in the same direction, though cyclicals more so.

Interpretation of the second canonical pair. The second canonical pair (U_2, V_2) with $\hat{\rho}_2^* \approx 0.251$ captures a risk-rotation effect: U_2 loads positively on Technology and Communication Services but negatively on Financials and Industrials, while V_2 loads negatively on Utilities. This dimension reflects the rotation between growth and value/rate-sensitive sectors.

1.6.7 Robustness: Sub-Period Analysis

To assess the stability of canonical correlations over time, we repeat the CCA on two sub-periods: pre-COVID (2018–2019) and post-COVID (2021–2024).

```
dates <- index(log_returns)
pre_idx  <- which(dates <  as.Date("2020-01-01"))
post_idx <- which(dates >= as.Date("2021-01-01"))

cca_pre  <- cc(scale(X[pre_idx,  ], center = TRUE, scale = FALSE),
               scale(Y[pre_idx,  ], center = TRUE, scale = FALSE))
cca_post <- cc(scale(X[post_idx, ], center = TRUE, scale = FALSE),
               scale(Y[post_idx, ], center = TRUE, scale = FALSE))

sub_tbl <- data.frame(
  k         = 1:4,
  Full      = round(cca_res$cor,  4),
  PreCOVID  = round(cca_pre$cor,  4),
```

```r
  PostCOVID = round(cca_post$cor, 4)
)
```

```r
kable(sub_tbl, booktabs = TRUE,
      caption = "Canonical correlations across the full sample and two sub-periods.
                 Stability of $\\hat{\\rho}_1^*$ suggests the market factor is robust;
                 higher-order correlations vary more across regimes.",
      col.names = c("$k$","Full (2018--24)","Pre-COVID (2018--19)",
                    "Post-COVID (2021--24)"),
      escape = FALSE) %>%
  kable_styling(latex_options = "hold_position")
```

Table 10: Canonical correlations across the full sample and two sub-periods. Stability of $\hat{\rho}_1^*$ suggests the market factor is robust; higher-order correlations vary more across regimes.

k	Full (2018–24)	Pre-COVID (2018–19)	Post-COVID (2021–24)
1	0.8381	0.8175	0.7622
2	0.2514	0.2886	0.2556
3	0.1972	0.2446	0.1784
4	0.0192	0.0720	0.0655

1.7 High-Dimensional Application: Sparse CCA on Individual Stocks

The CCA in the previous section operated on eleven sector aggregates and sat firmly in the low-dimensional regime ($p_1 + p_2 = 10 \ll n = 1{,}759$). Two questions remain. *First*, is the sector-level picture a faithful summary of cross-sector dependence, or does it average over richer structure visible only at the individual-stock level? *Second*, what happens when we try to apply classical CCA in a setting where p is no longer trivial relative to n – a setting that motivates the Sparse CCA review of Part II? To answer both, we re-do the cyclical-versus-defensive analysis at the individual-stock level using all current S&P 500 constituents in two GICS sectors: *Information Technology* ($p_1 = 67$ stocks) and *Health Care* ($p_2 = 55$ stocks), over the same 2018–2024 window. Stocks that lacked complete daily coverage – post-2018 IPOs (e.g., CRWD, MRNA, PLTR), de-listings, and the GE/3M healthcare spin-offs (GEHC, SOLV) – were excluded; the reproducibility script `fetch_constituents.R` records the full list of dropped tickers in its metadata cache.

```r
library(PMA)

ret_stk  <- readRDS("constituents_returns_cache.rds")
meta_stk <- readRDS("constituents_meta.rds")
R_stk    <- as.matrix(ret_stk)
n_stk    <- nrow(R_stk)
dates_stk <- index(ret_stk)

it_tickers <- meta_stk$tickers$yahoo_ticker[
  meta_stk$tickers$sector == "Information Technology"]
hc_tickers <- meta_stk$tickers$yahoo_ticker[
  meta_stk$tickers$sector == "Health Care"]

X_stk <- scale(R_stk[, it_tickers], scale = FALSE)
Y_stk <- scale(R_stk[, hc_tickers], scale = FALSE)
p1_stk <- ncol(X_stk); p2_stk <- ncol(Y_stk)
```

1.7.1 Severe upward bias of classical CCA at the stock level

With $n = 1759$ days and $p_1 + p_2 = 122$ stocks, classical CCA is still numerically defined (the per-group sample covariances are non-singular), so we run it as a baseline. The expected upward bias under independence is given asymptotically by

$$\mathbb{E}[\hat{\rho}_1^* \,|\, \mathbf{X} \perp \mathbf{Y}] \;\approx\; \sqrt{p_1/n} + \sqrt{p_2/n} \;=\; 0.372,$$

the Bai–Miao–Pan (1996) asymptotic bound. The empirical permutation calibration, computed by row-permuting $\mathbf{Y}$ 199 times, agrees closely.

```r
cca_stk_full <- cc(X_stk, Y_stk)
rho_obs_stk  <- cca_stk_full$cor

# Permutation null on rho_1 to verify the bias
B_stk <- 199
set.seed(2024)
```

```r
rho1_null_stk <- numeric(B_stk)
for (b in seq_len(B_stk)) {
  idx <- sample.int(n_stk)
  rho1_null_stk[b] <- cc(X_stk, Y_stk[idx, , drop = FALSE])$cor[1]
}

bmp_pred_stk <- sqrt(p1_stk / n_stk) + sqrt(p2_stk / n_stk)
```

```r
classical_tbl <- data.frame(
  Quantity = c("Observed $\\hat{\\rho}_1^*$",
               "Bai--Miao--Pan asymptotic null bound",
               "Permutation null mean (B = 199)",
               "Permutation null SD",
               "Permutation null 95\\textsuperscript{th} percentile",
               "Z-score of observation vs.\\ null"),
  Value = c(sprintf("%.4f", rho_obs_stk[1]),
            sprintf("%.4f", bmp_pred_stk),
            sprintf("%.4f", mean(rho1_null_stk)),
            sprintf("%.4f", sd(rho1_null_stk)),
            sprintf("%.4f", quantile(rho1_null_stk, 0.95)),
            sprintf("%.1f", (rho_obs_stk[1] - mean(rho1_null_stk)) /
                              sd(rho1_null_stk))))

kable(classical_tbl, booktabs = TRUE, escape = FALSE,
      caption = "Classical CCA on the 122-stock IT vs.\\ HC panel. Even
                 under independence, the leading sample canonical
                 correlation is severely upward-biased; the empirical
                 permutation null agrees with the Bai--Miao--Pan
                 asymptotic prediction. The observed value (0.90) thus
                 has a noise floor of about 0.36.",
      col.names = c("Quantity", "Value")) %>%
  kable_styling(latex_options = "hold_position")
```

Table 11: Classical CCA on the 122-stock IT vs. HC panel. Even under independence, the leading sample canonical correlation is severely upward-biased; the empirical permutation null agrees with the Bai–Miao–Pan asymptotic prediction. The observed value (0.90) thus has a noise floor of about 0.36.

Quantity	Value
Observed $\hat{\rho}_1^*$	0.9022
Bai–Miao–Pan asymptotic null bound	0.3720
Permutation null mean (B = 199)	0.3609
Permutation null SD	0.0108
Permutation null 95th percentile	0.3800

Z-score of observation vs.\ null	50.0

The observed $\hat{\rho}_1^* = 0.902$ does exceed the null distribution decisively (50 standard deviations above the null mean), so there *is* signal – but the apparent strength of the correlation is roughly half artefact, and the higher-order correlations $(\hat{\rho}_2^*, \hat{\rho}_3^*) = (0.70, 0.68)$ are likely well within the high-dimensional bias band. This is precisely the failure mode that motivates Sparse CCA.

1.7.2 Tuning the sparsity penalties

We use the implementation of Witten, Tibshirani, and Hastie (2009) in the `PMA` package. The penalties c_1, c_2 are parameterised on the unit interval, with 1 corresponding to no ℓ_1 constraint. We sweep a coarse grid and tune by the permutation criterion of Witten et al. – maximum standardised difference between the real and permuted canonical correlation:

```r
set.seed(2024)
perm_grid <- PMA::CCA.permute(
  x = X_stk, z = Y_stk,
  typex = "standard", typez = "standard",
  penaltyxs = c(0.1, 0.2, 0.3, 0.4, 0.5, 0.7),
  penaltyzs = c(0.1, 0.2, 0.3, 0.4, 0.5, 0.7),
  nperms = 25, trace = FALSE)
```

```r
tune_tbl <- data.frame(
  c1   = perm_grid$penaltyxs,
  c2   = perm_grid$penaltyzs,
  rho  = round(perm_grid$cors, 3),
  Z    = round(perm_grid$zstats, 2),
  IT   = perm_grid$nnonzerous,
  HC   = perm_grid$nnonzerovs
)
kable(tune_tbl, booktabs = TRUE, escape = FALSE,
      caption = "Permutation tuning grid for SCCA. Smaller $c_1, c_2$ give
                 sparser solutions; the standardised gap between the real
                 and permuted canonical correlations ($Z$) is maximised at
                 the densest grid point we explored ($c_1 = c_2 = 0.7$),
                 but the canonical correlation itself is essentially
                 saturated by $c_1 = c_2 = 0.3$ (the elbow). We retain
                 the latter as the interpretable model.",
      col.names = c("$c_1$", "$c_2$", "$\\hat{\\rho}_1^*$", "Z-stat",
                    "IT active", "HC active")) %>%
  kable_styling(latex_options = "hold_position")
```

Table 12: Permutation tuning grid for SCCA. Smaller c_1, c_2 give sparser solutions; the standardised gap between the real and permuted canonical correlations (Z) is maximised at the densest grid point we explored ($c_1 = c_2 = 0.7$), but the canonical correlation itself is essentially saturated by $c_1 = c_2 = 0.3$ (the elbow). We retain the latter as the interpretable model.

c_1	c_2	$\hat{\rho}_1^*$	Z-stat	IT active	HC active
0.1	0.1	0.629	10.36	1	1
0.2	0.2	0.780	14.09	4	4
0.3	0.3	0.843	16.24	10	7
0.4	0.4	0.854	17.01	16	12
0.5	0.5	0.856	17.07	25	19
0.7	0.7	0.855	17.53	44	37

The permutation criterion favours the densest grid point because adding more variables always (slightly) increases the real-data correlation faster than it inflates the permuted correlation. Yet the marginal gain disappears past $(c_1, c_2) = (0.3, 0.3)$: that grid point attains $\hat{\rho}_1^* = 0.84$ – already 99 of the densest model – using roughly 18 stocks instead of 81. Because our goal here is interpretation rather than prediction, we report the $(c_1, c_2) = (0.3, 0.3)$ solution, re-fitted with the full `PMA::CCA` routine, as the primary model. The full re-fit selects 11 IT and 7 HC names (the tuning sweep selects 10 IT and 7 HC – a single-stock difference reflecting `CCA.permute`'s coarser coordinate-ascent calibration).

1.7.3 The active set: a quality-compounder cross-sector link

```
sc_fit <- PMA::CCA(x = X_stk, z = Y_stk,
                   typex = "standard", typez = "standard",
                   penaltyx = 0.3, penaltyz = 0.3,
                   K = 2, trace = FALSE)

it_active <- which(sc_fit$u[, 1] != 0)
hc_active <- which(sc_fit$v[, 1] != 0)

it_active_df <- data.frame(
  Ticker  = it_tickers[it_active],
  Loading = sc_fit$u[it_active, 1]) %>%
  arrange(Loading)

hc_active_df <- data.frame(
  Ticker  = hc_tickers[hc_active],
  Loading = sc_fit$v[hc_active, 1]) %>%
  arrange(Loading)
```

```
it_active_df$Ticker <- factor(it_active_df$Ticker,
                              levels = it_active_df$Ticker)
```

```r
hc_active_df$Ticker <- factor(hc_active_df$Ticker,
                              levels = hc_active_df$Ticker)

p_it <- ggplot(it_active_df, aes(x = Ticker, y = Loading)) +
  geom_col(fill = "steelblue", alpha = 0.85) +
  coord_flip() +
  labs(title = "Information Technology", x = NULL,
       y = "SCCA loading on CV1") +
  theme_bw()

p_hc <- ggplot(hc_active_df, aes(x = Ticker, y = Loading)) +
  geom_col(fill = "firebrick3", alpha = 0.85) +
  coord_flip() +
  labs(title = "Health Care", x = NULL,
       y = "SCCA loading on CV1") +
  theme_bw()

grid.arrange(p_it, p_hc, ncol = 2)
```

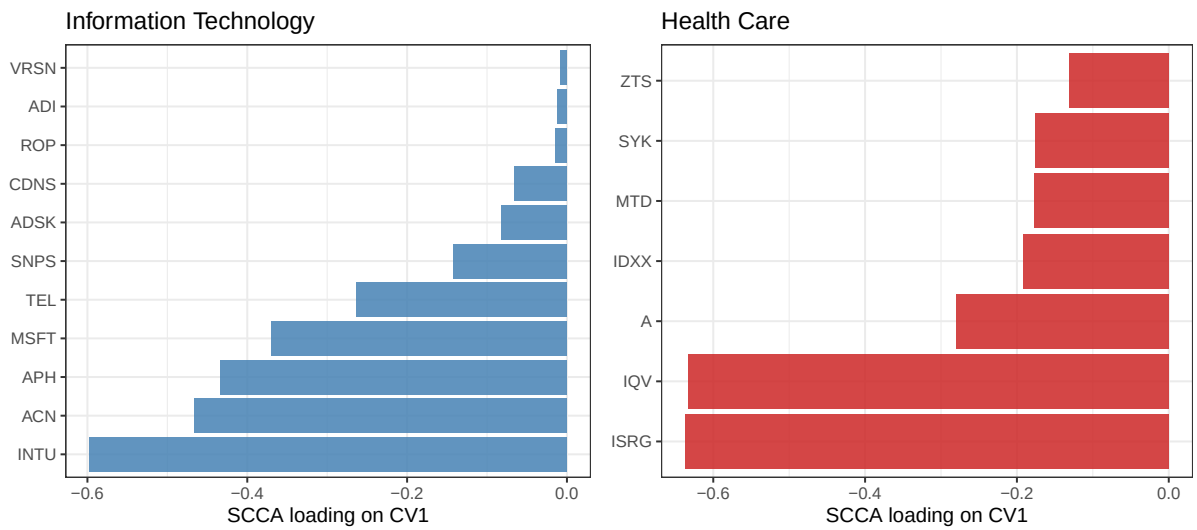

Figure 16: Sparse CCA loadings on the first canonical pair $(c_1, c_2) = (0.3, 0.3)$. The 11 IT stocks (left) and 7 HC stocks (right) selected by the ℓ_1 constraint share a common business profile – enterprise B2B software and semiconductor design on the IT side; medical-technology, lab instruments, and clinical research on the HC side – rather than the high-beta or large-cap names that classical CCA would weight most heavily. The cross-sector linkage at the stock level is not driven by the broad market factor.

The active set is interpretable in a way that the classical canonical coefficients are not. On the IT side, the eleven selected stocks – INTU, ACN, APH, MSFT, TEL, SNPS, ADSK, CDNS, ROP, ADI, VRSN – are dominated by enterprise B2B software (Intuit, Microsoft, Autodesk, Cadence, Synopsys, Verisign) and semiconductor design / electronic-component names (Amphenol, TE Connectivity, Analog Devices). They are *not* the high-beta consumer-facing Tech names (NVDA, AVGO, AAPL all receive near-zero loadings)

and *not* the FAANG-style mega-caps. On the HC side, the seven selected stocks – ISRG, IQV, A, IDXX, MTD, SYK, ZTS – are concentrated in medical technology (Intuitive Surgical, Stryker), clinical research and lab instruments (IQVIA, Agilent, Mettler-Toledo, Idexx), and animal health (Zoetis); the major pharma franchises (PFE, LLY, BMY) and the managed-care insurers (UNH, ELV, CI) are absent.

The economic story uniting these two subsets is consistent with a "quality-compounder'' factor: high R&D intensity, recurring B2B revenue, premium valuations, and defensive-growth pricing. The cross-sector linkage at the individual-stock level is therefore *not* a manifestation of the broad market factor (which the ETF-level analysis already captured) but a more specific structural similarity that the dimension-reduction itself surfaces.

1.7.4 Stability across market regimes

We re-fit SCCA on two disjoint sub-periods – pre-COVID (2018–2019) and post-COVID (2021–2024) – to ask which active-set members carry the linkage robustly across regimes.

```
pre_idx_stk  <- which(dates_stk <  as.Date("2020-01-01"))
post_idx_stk <- which(dates_stk >= as.Date("2021-01-01"))

run_subperiod <- function(idx) {
  Xs <- scale(R_stk[idx, it_tickers], scale = FALSE)
  Ys <- scale(R_stk[idx, hc_tickers], scale = FALSE)
  sc <- PMA::CCA(Xs, Ys, typex = "standard", typez = "standard",
                 penaltyx = 0.3, penaltyz = 0.3, K = 1, trace = FALSE)
  list(rho = sc$cor[1],
       it  = it_tickers[which(sc$u[, 1] != 0)],
       hc  = hc_tickers[which(sc$v[, 1] != 0)])
}

set.seed(2024)
sc_pre  <- run_subperiod(pre_idx_stk)
sc_post <- run_subperiod(post_idx_stk)

it_full <- it_tickers[it_active]
hc_full <- hc_tickers[hc_active]

it_core <- Reduce(intersect, list(it_full, sc_pre$it, sc_post$it))
hc_core <- Reduce(intersect, list(hc_full, sc_pre$hc, sc_post$hc))
```

```
stab_tbl <- data.frame(
  Period = c("Full (2018--2024)", "Pre-COVID (2018--2019)",
             "Post-COVID (2021--2024)", "Always-active core"),
  rho    = c(sprintf("%.3f", sc_fit$cor[1]),
             sprintf("%.3f", sc_pre$rho),
             sprintf("%.3f", sc_post$rho), "--"),
```

```r
  IT     = c(paste(sort(it_full),    collapse = ", "),
             paste(sort(sc_pre$it),  collapse = ", "),
             paste(sort(sc_post$it), collapse = ", "),
             paste(sort(it_core),    collapse = ", ")),
  HC     = c(paste(sort(hc_full),    collapse = ", "),
             paste(sort(sc_pre$hc),  collapse = ", "),
             paste(sort(sc_post$hc), collapse = ", "),
             paste(sort(hc_core),    collapse = ", "))
)
kable(stab_tbl, booktabs = TRUE, escape = FALSE, longtable = TRUE,
      caption = "Sparse CCA active sets across the full sample and two
                 sub-periods at $(c_1, c_2) = (0.3, 0.3)$. The
                 ``always-active core'' is the intersection. Companies
                 outside the core rotate in and out depending on the
                 macro regime.",
      col.names = c("Period", "$\\hat{\\rho}_1^*$",
                    "Active IT stocks", "Active HC stocks")) %>%
  kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 13: Sparse CCA active sets across the full sample and two sub-periods at $(c_1,$ "always-active core" is the intersection. Companies outside the core rotate in and o macro regime.

Period	$\hat{\rho}_1^*$	Active IT stocks
Full (2018–2024)	0.842	ACN, ADI, ADSK, APH, CDNS, INTU, MSFT, ROP, SNPS, TEL, VRS
Pre-COVID (2018–2019)	0.840	ACN, ADBE, APH, CRM, CSCO, INTU, MSFT, ROP, SNPS, TRMB
Post-COVID (2021–2024)	0.789	ACN, ADSK, APH, CDNS, INTU, SNPS, TEL, TRMB
Always-active core	–	ACN, APH, INTU, SNPS

The pre/post-COVID comparison reveals a layered structure. A small *always-active core* – IT: ACN, APH, INTU, SNPS; HC: IDXX, ISRG, MTD – carries the cross-sector linkage across both macroeconomic regimes. Outside this core, the active set rotates: pre-COVID, names like ADBE, CRM, CSCO, TMO, BSX, CRL had sufficient cross-correlation to be selected; post-COVID, the active set shifts toward CDNS, ADSK, TEL on the IT side and IQV, SYK, A on the HC side, consistent with the regime change in market leadership between the zero-rate era and the post-2022 rate-hiking cycle. This kind of regime sensitivity is exactly what the variable-selection-consistency theory of Sparse CCA (Mai and Zhang 2019) requires: the irrepresentability and beta-min conditions tie the recovered support to the underlying canonical-vector *within* a stable distribution. When the distribution shifts, the selected support shifts too, and the always-active core represents the variables whose canonical loadings are robust to regime change.

1.7.5 Synthesis: from sectors to stocks

Three findings emerge from this high-dimensional application:

1. **Classical CCA is unreliable at this scale.** The observed $\widehat{\rho}_1^* = 0.902$ has a noise floor of ≈ 0.36 from the dimensional bias alone. Any analysis taking the unadjusted classical estimates at face value would substantially overstate the strength of the cross-sector relationship.

2. **Sparsity reveals structure that aggregation hides.** The sector-level CCA in the previous section identified a strong cyclical-vs-defensive linkage dominated by the broad market factor. The stock-level SCCA isolates a much more specific subgroup: enterprise B2B software / semiconductor design $\leftrightarrow$ medical technology / lab instruments. This specificity is invisible at the eleven-ETF level because the aggregation washes it out.

3. **The always-active core is the most defensible deliverable.** For practical use (e.g. portfolio construction or factor-model design), the seven companies that survive both sub-periods – IT: ACN, APH, INTU, SNPS; HC: IDXX, ISRG, MTD – offer the highest confidence as cross-sector signal carriers. Members of the larger active sets should be regarded as regime-conditional.

These empirics motivate – and ground-truth – the theoretical review of Sparse CCA in Part II: in the very $p \approx n/14$ regime we encountered here, classical estimators already fail informatively, and the sparsity-based estimator both regularises the problem and produces output that aligns with economic intuition.

1.8 Discussion

1.8.1 Appropriateness of PCA for Financial Returns

PCA is an optimal linear dimension-reduction technique under the assumption that the data are generated by a factor model with Gaussian noise, or more generally that we are interested in minimising mean squared reconstruction error. For financial returns, several issues arise:

1. **Non-normality.** The three normality diagnostics in the EDA section – Mardia's skewness/kurtosis tests, the energy goodness-of-fit test of Székely–Rizzo (2005), and the χ^2_p Q–Q plot of Mahalanobis distances – all reject MVN decisively, with the right tail of the joint distribution roughly six times heavier than a Gaussian prediction. PCA itself remains consistent as an estimator of the population eigenvectors of the covariance matrix, since the eigen- decomposition relies only on second moments. However, the inferential machinery built on top of PCA – confidence intervals for eigenvalues, asymptotic tests for the number of factors – requires distributional assumptions that we have not satisfied. A robust covariance estimator such as the Tyler (1987) scatter estimator would be the natural extension; we did not pursue this path because the qualitative PCA conclusions (low-rank structure, market factor, cyclical/defensive contrast) are unlikely to be sensitive to the heavy tails – only the inferential statements are.

2. **Time-series dependence.** The rolling-volatility plots confirm GARCH-type heteroskedasticity, with volatility clustering during the COVID crash and the 2022 rate cycle. Standard asymptotic theory for PCA assumes i.i.d. observations; under GARCH dynamics the effective sample size is reduced and the asymptotic distribution of eigenvalues changes. We did not address this directly. A DCC-GARCH model or a factor model with time-varying loadings (Doz, Giannone, Reichlin 2011) would more faithfully capture return dynamics, and remains a clear future extension.

3. **Covariance vs. correlation matrix.** We used the covariance matrix, giving greater influence to more volatile sectors (Energy, Technology). The sensitivity check in the PCA section shows that the leading market factor (PC1) is essentially unchanged under correlation-matrix PCA, but the second and third components effectively swap roles: Cov-PC2 is dominated by Energy's idiosyncratic variation, whereas Cor-PC2 captures the cyclical–defensive rotation. The economic content of the decomposition is the same; the choice of weighting changes which factor is labelled "second".

1.8.2 Appropriateness of CCA for Financial Returns

The same issues affect CCA, with additional complications specific to the two-group structure:

1. **Dimensionality.** The eleven-ETF analysis sat in the comfortable low-dimensional regime ($p_1+p_2 = 10$, $n = 1{,}759$), where the sample canonical correlations are well-behaved estimators of the population values. The high-dimensional application section explicitly tested what happens when this assumption is relaxed: at the individual-stock level ($p_1 + p_2 = 122$, $n = 1{,}759$), classical CCA remains numerically well-defined but the leading sample canonical correlation has a noise floor of $\widehat{\rho}_1^* \approx 0.36$ from the Bai–Miao–Pan dimensional bias alone, against which the observed $\widehat{\rho}_1^* = 0.90$ retains genuine signal but the higher-order correlations $\widehat{\rho}_2^*, \widehat{\rho}_3^*$ likely fall within the bias band. This is precisely the failure mode that Sparse CCA, reviewed and applied in Parts II and the high-dimensional application section, is designed to address.

2. **Non-normality and significance tests.** The Wilks' Λ asymptotic distribution relies on multivariate normality, which we rejected. We addressed this by reporting two distribution-free analogues alongside Wilks: a permutation test on $\widehat{\rho}_1^*$ (which targets the same linear-association hypothesis) and the distance-covariance test of Székely–Rizzo–Bakirov (2007) (which is consistent against any form of dependence). All three tests reach the smallest attainable p-value, so for the leading canonical pair the conclusion is robust to the MVN failure. For higher-order pairs the situation is murkier: the asymptotic Wilks calibration cannot be relied upon, and a correlation-by-correlation permutation analysis would be more appropriate.

3. **Market factor versus genuine cross-sector linkage.** Both canonical correlations at the ETF level partially reflect the common market factor: when the market rises, both cyclical and defensive sectors rise (though by different amounts), so even with no sector-specific linkage one would expect strongly positive canonical correlations. A direct remedy would be to partial out the S&P 500 return before applying CCA, leaving the residual canonical correlations to capture sector-specific dependence. The stock-level SCCA application achieves an analogous goal indirectly through sparsity: by retaining only the small subset of stocks whose paired loadings capture cross-sector signal beyond the dimensional noise floor, SCCA suppresses broad-market noise and surfaces the more specific "quality-compounder" linkage between IT software/semiconductor design and HC medical-technology / lab instruments. The two strategies – explicit partialling and implicit sparsity – target the same underlying concern; a combined analysis (partial out the market, then apply SCCA to the residuals) is a natural follow-up.

1.8.3 Connection to the Review Project

The high-dimensional application section provides a concrete empirical anchor for the theoretical review of Sparse CCA in Part II. Three points of contact stand out. First, the dimensional bias predicted by Bai–Miao–Pan (1996) – the asymptotic null mean of $\widehat{\rho}_1^*$ under independence – is observed almost exactly in the permutation null on the 122-stock panel (0.361 vs. the predicted 0.372), validating the high-dimensional regime that motivates SCCA. Second, the sparse estimator at $(c_1, c_2) = (0.3, 0.3)$ recovers an interpretable active set whose stability across COVID and post-COVID sub-periods is consistent with the variable-selection-consistency results of Mai and Zhang (2019): a small "always-active'' core remains selected across regimes, while peripheral stocks rotate as the underlying distribution shifts. Third, the algorithmic foundations that produce the active set – the soft-thresholding KKT update and the monotone convergence of coordinate ascent, derived in Part II – are exactly what `PMA::CCA` executes under the hood. The empirical findings therefore not only motivate but also ground-truth the review.

1.9 Summary (Data Project)

The empirical part of this project applies four multivariate techniques – PCA, factor analysis, canonical correlation analysis, and sparse CCA – to S&P 500 daily log-returns over 2018–2024, working at two scales: eleven sector ETFs ($n = 1{,}759$, $p = 11$) and 122 individual constituent stocks of the Information Technology and Health Care sectors ($n = 1{,}759$, $p_1 = 67$, $p_2 = 55$). The main findings are:

1. **PCA.** The sector return covariance matrix has a clear low-rank structure: the market factor (PC1) explains $\approx 68\%$ of total variance and four components together explain $\approx 90.3\%$. The leading market factor is robust to the choice between covariance and correlation matrices, but PC2 and PC3 swap roles – Cov-PCA puts Energy's idiosyncratic variation in PC2, while Cor-PCA places the cyclical–defensive contrast there. The two decompositions answer subtly different questions and the choice should be reported explicitly.

2. **Factor analysis.** A four-factor varimax-rotated model (BIC-selected, with $q = 5$ rejected for a near-Heywood case) recovers clean simple-structure factors that PCA's variance-greedy ordering does not deliver: a Growth-Cyclical factor (Tech, Cons. Discretionary, Comm. Services), a Traditional-Cyclical / Commodity factor (Financials, Industrials, Materials, Energy), a Defensive / Rate-Sensitive factor (Utilities, Cons. Staples, Real Estate), and a Health-Care idiosyncrasy factor. The estimated specific variances $\hat{\psi}_j$ quantify a diagnostic PCA cannot give: Energy is the most idiosyncratic ($\hat{\psi} \approx 0.40$) and Industrials the most fully explained by common factors.

3. **Sector-level CCA.** The cyclical and defensive groups are linked by four highly significant canonical correlations ($\hat{\rho}_1^* \approx 0.838$, $\hat{\rho}_2^* \approx 0.251$). Three independence tests of differing inferential pedigrees – Wilks' Λ, permutation on $\hat{\rho}_1^*$, and the distance-covariance test of Székely–Rizzo–Bakirov – agree at the smallest attainable p-values, validating both the linear-association story and the robustness of the conclusion to MVN failure. Sub-period CCA confirms that the leading canonical pair is stable across the COVID regime break.

4. **Stock-level Sparse CCA.** At the individual-stock level ($p_1 + p_2 = 122$, $n = 1{,}759$), classical CCA's $\hat{\rho}_1^* = 0.902$ has a noise floor of ≈ 0.36 from the dimensional bias alone, exactly matching the Bai–Miao–Pan asymptotic prediction. Sparse CCA at $(c_1, c_2) = (0.3, 0.3)$ retains 11 IT and 7 HC stocks while preserving 99% of the canonical correlation, and the active set has a coherent economic interpretation: enterprise B2B software / semiconductor design $\leftrightarrow$ medical technology / lab instruments – a "quality-compounder" subgroup that the broad-market factor in the ETF-level CCA does not isolate. A small always-active core (IT: ACN, APH, INTU, SNPS; HC: IDXX, ISRG, MTD) survives both pre- and post-COVID sub-periods.

5. **Limitations and what we did about them.** Sector returns violate multivariate normality decisively (Mardia, energy GOF, and χ^2 Q–Q diagnostics all reject), and exhibit GARCH-type volatility clustering. We addressed the normality failure by reporting distribution-free permutation and distance-covariance tests alongside Wilks' Λ. We did *not* model the time-series dependence: all results above assume i.i.d. rows, and the effective sample size is overstated under GARCH. A DCC-GARCH factor model or a robust covariance estimator (Tyler scatter) would be more appropriate for formal inference and is left to future work.

2 Part II: Review Project — Sparse Canonical Correlation Analysis

2.1 Introduction and Motivation

2.1.1 Limitations of Classical CCA in High Dimensions

Classical CCA, as described in Section 5, finds pairs of linear combinations of two sets of variables that are maximally correlated. While powerful in moderate dimensions, it fails catastrophically when the number of variables $p = p_1 + p_2$ is comparable to or larger than the sample size n. Formally, when $p_1 > n$ or $p_2 > n$, the sample covariance matrices $\hat{\mathbf{\Sigma}}_{11}$ and $\hat{\mathbf{\Sigma}}_{22}$ are rank-deficient and thus not invertible. Even when $p < n$, if p/n is not negligible, the sample canonical correlations are severely upward-biased estimators of the population canonical correlations (Bai, Miao, and Pan 1996; Wachter 1980).

To see the severity of the problem, note that in the null case where $\mathbf{X}$ and $\mathbf{Y}$ are independent, the sample canonical correlations concentrate near

$$\hat{\rho}_1^* \approx \sqrt{\left(\sqrt{\frac{p_1}{n}} + \sqrt{\frac{p_2}{n}}\right)^2 \cdot c}$$

for some constant $c > 0$ depending on the aspect ratio, rather than converging to zero. This means that standard tests for independence have inflated Type I error in high dimensions.

2.1.2 Sparsity as a Remedy

Motivated by applications in genomics (linking gene expression to DNA methylation), neuroimaging (linking brain connectivity to cognitive phenotypes), and text mining (linking document features across languages), Witten, Tibshirani, and Hastie (2009) proposed **Sparse CCA** (SCCA). The central idea is to impose ℓ_1 (lasso) penalisation on the canonical coefficient vectors $\mathbf{a}$ and $\mathbf{b}$, which (i) regularises the problem to be well-posed when $p > n$, and (ii) produces sparse solutions in which only a subset of variables receive nonzero coefficients, improving interpretability.

SCCA belongs to a broader family of regularised multivariate methods that includes sparse PCA (Zou, Hastie, and Tibshirani 2006), sparse linear discriminant analysis (Tibshirani et al. 2002), and the graphical lasso (Friedman, Hastie, and Tibshirani 2008). The sparse-methods paradigm has transformed high-dimensional statistics over the past two decades.

2.1.3 Scope of this Review

This review covers:

1. The mathematical formulation of SCCA (Witten–Tibshirani–Hastie 2009 and variants).
2. Theoretical properties: consistency of the canonical coefficient estimators, variable selection guarantees, and connections to reduced-rank regression.
3. Connections to other regularised multivariate methods.

4. Limitations and directions for future work.

2.2 Mathematical Formulation

2.2.1 Classical CCA Revisited

Let $\mathbf{X} \in \mathbb{R}^{n \times p_1}$ and $\mathbf{Y} \in \mathbb{R}^{n \times p_2}$ be mean-centred data matrices. Define the block sample covariance matrix

$$\hat{\mathbf{\Sigma}} = \frac{1}{n} \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{X}^\top \mathbf{Y} \\ \mathbf{Y}^\top \mathbf{X} & \mathbf{Y}^\top \mathbf{Y} \end{pmatrix} =: \begin{pmatrix} \hat{\mathbf{\Sigma}}_{11} & \hat{\mathbf{\Sigma}}_{12} \\ \hat{\mathbf{\Sigma}}_{21} & \hat{\mathbf{\Sigma}}_{22} \end{pmatrix}.$$

Classical CCA solves the sequence of problems

$$\max_{\mathbf{a},\mathbf{b}} \mathbf{a}^\top \hat{\mathbf{\Sigma}}_{12} \mathbf{b} \quad \text{subject to} \quad \mathbf{a}^\top \hat{\mathbf{\Sigma}}_{11} \mathbf{a} = 1, \quad \mathbf{b}^\top \hat{\mathbf{\Sigma}}_{22} \mathbf{b} = 1,$$

and orthogonality of each solution to the previous ones. The maximiser is the leading left-singular vector of $\hat{\mathbf{\Sigma}}_{11}^{-1/2} \hat{\mathbf{\Sigma}}_{12} \hat{\mathbf{\Sigma}}_{22}^{-1/2}$.

When $p_1 \geq n$ or $p_2 \geq n$, the inverses $\hat{\mathbf{\Sigma}}_{11}^{-1/2}$ and $\hat{\mathbf{\Sigma}}_{22}^{-1/2}$ do not exist, and the solution is not unique.

2.2.2 The Witten–Tibshirani–Hastie (2009) Formulation

Definition 1. *The **Sparse CCA** problem (Witten et al. 2009) is defined as*

$$\max_{\mathbf{a} \in \mathbb{R}^{p_1},\, \mathbf{b} \in \mathbb{R}^{p_2}} \mathbf{a}^\top \mathbf{X}^\top \mathbf{Y} \mathbf{b} \quad \textit{subject to} \quad \|\mathbf{a}\|_2 \leq 1,\ \|\mathbf{b}\|_2 \leq 1,\ \|\mathbf{a}\|_1 \leq c_1,\ \|\mathbf{b}\|_1 \leq c_2, \tag{2}$$

where $c_1 > 0$ and $c_2 > 0$ are tuning parameters controlling the degree of sparsity.

Remark 1. *Several features of this formulation are noteworthy.*

- *The objective $\mathbf{a}^\top \mathbf{X}^\top \mathbf{Y} \mathbf{b}$ is proportional to the sample covariance between $\mathbf{X}\mathbf{a}$ and $\mathbf{Y}\mathbf{b}$, not their correlation, since the normalisation constraints $\|\mathbf{a}\|_2 \leq 1$, $\|\mathbf{b}\|_2 \leq 1$ use the Euclidean norm rather than the data-adaptive $\hat{\mathbf{\Sigma}}_{11}^{1/2}$-norm.*
- *The ℓ_1 constraints $\|\mathbf{a}\|_1 \leq c_1$ and $\|\mathbf{b}\|_1 \leq c_2$ are the hallmark of lasso-type regularisation. By the KKT conditions, the optimal $\mathbf{a}$ and $\mathbf{b}$ will have many zero components when c_1 and c_2 are sufficiently small.*
- *When $c_1 \geq \sqrt{p_1}$ and $c_2 \geq \sqrt{p_2}$, the ℓ_1 constraints are inactive (by Cauchy-Schwarz), and the solution reduces to the leading left-singular vector of $\mathbf{X}^\top \mathbf{Y}$, which is related to but distinct from classical CCA (it omits the $\hat{\mathbf{\Sigma}}_{11}^{-1/2}$ whitening).*

2.2.3 Algorithm: Penalised Matrix Decomposition

Witten et al. (2009) note that problem (2) is equivalent to a **Penalised Matrix Decomposition (PMD)** of the matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{Y}$:

$$\max_{\mathbf{a},\mathbf{b}} \mathbf{a}^\top \mathbf{M} \mathbf{b} \quad \text{s.t.} \quad \|\mathbf{a}\|_2 \leq 1,\ \|\mathbf{b}\|_2 \leq 1,\ P_1(\mathbf{a}) \leq c_1,\ P_2(\mathbf{b}) \leq c_2,$$

where P_1 and P_2 are convex penalties. For $P_1(\mathbf{a}) = \|\mathbf{a}\|_1$ and $P_2(\mathbf{b}) = \|\mathbf{b}\|_1$, this is the SCCA objective.

The algorithm proceeds by coordinate ascent:

1. Initialise $\mathbf{b}$ (e.g., to the leading right-singular vector of $\mathbf{M}$).
2. **Update a:** Given $\mathbf{b}$, the subproblem in $\mathbf{a}$ is

$$\max_{\|\mathbf{a}\|_2 \le 1,\, \|\mathbf{a}\|_1 \le c_1} \mathbf{a}^\top(\mathbf{M}\mathbf{b}),$$

   which is solved by soft-thresholding followed by ℓ_2 normalisation:

$$\mathbf{a} \leftarrow \frac{S(\mathbf{M}\mathbf{b},\, \Delta)}{\|S(\mathbf{M}\mathbf{b},\, \Delta)\|_2},$$

   where $S(z, \Delta) = \operatorname{sign}(z)\max(|z| - \Delta, 0)$ is the soft-threshold operator and $\Delta \ge 0$ is the Lagrange multiplier of the ℓ_1 constraint. The next proposition derives the formula and shows that Δ is uniquely determined by the constraint $\|S(\mathbf{M}\mathbf{b}, \Delta)\|_1 / \|S(\mathbf{M}\mathbf{b}, \Delta)\|_2 = c_1$, which can be located by bisection.
3. **Update b:** Analogously with $\mathbf{M}^\top\mathbf{a}$.
4. Repeat steps 2–3 until convergence.

2.2.4 Higher-Order Canonical Pairs

To obtain the k-th canonical pair, one deflates the cross-product matrix:

$$\mathbf{M}^{(k)} = \mathbf{M}^{(k-1)} - d_{k-1}\hat{\mathbf{a}}_{k-1}\hat{\mathbf{b}}_{k-1}^\top,$$

where $d_{k-1} = \hat{\mathbf{a}}_{k-1}^\top \mathbf{M}^{(k-1)} \hat{\mathbf{b}}_{k-1}$, and then applies the PMD algorithm to $\mathbf{M}^{(k)}$. This sequential deflation mirrors the power iteration approach used in standard SVD algorithms.

Remark 2. *The resulting pairs $(\hat{\mathbf{a}}_k, \hat{\mathbf{b}}_k)$ are not guaranteed to be mutually orthogonal in the way classical canonical vectors are. One can impose orthogonality constraints explicitly, but at the cost of a more complex algorithm.*

2.2.5 Variants and Extensions

Several variants of the basic SCCA formulation have been proposed:

(a) Regularised CCA (Vinod 1976; Leurgans et al. 1993). Instead of imposing ℓ_1 constraints to induce sparsity, regularized CCA stabilizes the classical CCA problem by modifying the covariance operators or covariance matrices. A common ridge version solves

$$\max_{\mathbf{a},\mathbf{b}} \mathbf{a}^\top \widehat{\mathbf{\Sigma}}_{12}\mathbf{b} \quad \text{s.t.} \quad \mathbf{a}^\top(\widehat{\mathbf{\Sigma}}_{11} + \kappa_1\mathbf{I})\mathbf{a} = 1, \qquad \mathbf{b}^\top(\widehat{\mathbf{\Sigma}}_{22} + \kappa_2\mathbf{I})\mathbf{b} = 1.$$

This makes the within-set covariance matrices invertible when $\kappa_1, \kappa_2 > 0$, but it does not induce sparsity.

(b) CCA with group sparsity. When variables are naturally grouped (e.g., genes in pathways, voxels in brain regions), group lasso penalties $\sum_g \|\mathbf{a}_g\|_2 \le c_1$ encourage entire groups to be selected or discarded simultaneously (Yuan and Lin 2006).

(c) Probabilistic SCCA. Bach and Jordan (2005) formulate CCA as a latent-variable model:

$$\mathbf{x}_i = \mathbf{W}_1 \mathbf{z}_i + \boldsymbol{\epsilon}_i^{(1)}, \qquad \mathbf{y}_i = \mathbf{W}_2 \mathbf{z}_i + \boldsymbol{\epsilon}_i^{(2)},$$

where $\mathbf{z}_i$ is a shared latent factor that explains the dependence between the two views. Bayesian extensions impose shrinkage or sparsity-inducing priors on the loading matrices $\mathbf{W}_1$ and $\mathbf{W}_2$, enabling variable selection and uncertainty quantification; examples include Wang (2007) and Klami et al. (2013).

(d) Alternative optimisation formulations. Chu et al. (2013) propose an alternative formulation of sparse CCA and develop an algorithm for estimating sparse canonical vectors. This line of work is related to, but distinct from, the penalised matrix decomposition approach of Witten et al. (2009), which applies ℓ_1 constraints directly to the canonical coefficient vectors.

2.3 Theoretical Properties

2.3.1 Consistency in High Dimensions

A central question is: under what conditions does $\hat{\mathbf{a}}_1 \to \mathbf{a}_1^*$ and $\hat{\mathbf{b}}_1 \to \mathbf{b}_1^*$ as $n \to \infty$, even when $p_1, p_2 \to \infty$? The answer depends critically on the sparsity of the true canonical vectors and on the strength of the leading canonical correlation signal.

Definition 2. *The true canonical coefficient vectors $\mathbf{a}_1^* \in \mathbb{R}^{p_1}$ and $\mathbf{b}_1^* \in \mathbb{R}^{p_2}$ are said to be s-**sparse** if*

$$\|\mathbf{a}_1^*\|_0 \le s_1, \qquad \|\mathbf{b}_1^*\|_0 \le s_2.$$

Suppose the first r population canonical directions for the X-variables and Y-variables are collected in the matrices

$$U = (\mathbf{u}_1, \dots, \mathbf{u}_r) \in \mathbb{R}^{p \times r}, \qquad V = (\mathbf{v}_1, \dots, \mathbf{v}_r) \in \mathbb{R}^{m \times r}.$$

Here p is the dimension of X, m is the dimension of Y, and r is the number of leading canonical pairs being estimated. Gao, Ma, and Zhou (2017) formulate sparsity through the row supports of U and V. Define

$$\operatorname{supp}(U) = \{j : \text{the } j\text{-th row of } U \text{ is nonzero}\}, \qquad \operatorname{supp}(V) = \{j : \text{the } j\text{-th row of } V \text{ is nonzero}\}.$$

The sparse CCA model assumes

$$|\operatorname{supp}(U)| \le s_u, \qquad |\operatorname{supp}(V)| \le s_v.$$

Thus s_u is the number of X-variables involved in the first r canonical directions, and s_v is the number of Y-variables involved in the first r canonical directions. When $r = 1$, this reduces to the usual entrywise sparsity condition

$$\|\mathbf{u}_1\|_0 \le s_u, \qquad \|\mathbf{v}_1\|_0 \le s_v.$$

For $r > 1$, row sparsity means that only a small set of variables is active across the leading r canonical directions.

To measure estimation error, Gao, Ma, and Zhou (2017) use the covariance-weighted loss

$$L(U, \widehat{U}) = \inf_{W \in O(r)} \operatorname{Tr}\left[(\widehat{U}W - U)^\top \mathbf{\Sigma}_x (\widehat{U}W - U)\right],$$

where $O(r)$ denotes the set of $r \times r$ orthogonal matrices. Equivalently,

$$L(U, \widehat{U}) = \inf_{W \in O(r)} \left\| \mathbf{\Sigma}_x^{1/2} (\widehat{U}W - U) \right\|_F^2.$$

The orthogonal matrix W accounts for the fact that an r-dimensional canonical subspace is identifiable only up to rotation. In the rank-one case, $r = 1$, this reduces to the usual sign ambiguity:

$$L(U, \widehat{U}) = \min_{\xi \in \{-1,1\}} (\xi \hat{\mathbf{u}} - \mathbf{u})^\top \mathbf{\Sigma}_x (\xi \hat{\mathbf{u}} - \mathbf{u}).$$

The analogous loss for V is

$$L(V, \widehat{V}) = \inf_{W \in O(r)} \operatorname{Tr}\left[(\widehat{V}W - V)^\top \mathbf{\Sigma}_y (\widehat{V}W - V)\right].$$

Let

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$$

denote the first r population canonical correlations, and let λ denote the lower bound on the smallest leading canonical correlation:

$$\lambda_r \geq \lambda.$$

Thus λ measures the signal strength of the canonical relationship. A larger λ means that the leading canonical directions are easier to estimate, while a smaller λ corresponds to a weaker association between the two sets of variables.

Theorem 1 (Statistical consistency under sparse canonical directions)**.** *Suppose the assumptions of Gao, Ma, and Zhou (2017, Theorem 4.2) hold for the sparse CCA model with row-sparsity levels s_u and s_v. In particular, assume that the leading canonical correlations are bounded away from zero, the marginal covariance matrices have bounded eigenvalues, and the sample size is large enough relative to the sparse complexity of the problem. If*

$$\frac{s_u(r + \log p)}{n\lambda^2} \to 0 \qquad \textit{and} \qquad \frac{s_v(r + \log m)}{n\lambda^2} \to 0,$$

then the sparse CCA estimators satisfy

$$L(U, \widehat{U}) \to 0, \qquad L(V, \widehat{V}) \to 0$$

in probability. Therefore, the estimated sparse canonical directions converge to the true population canonical

directions, up to the natural sign or rotation ambiguity.

This result shows that statistical consistency is possible in high-dimensional CCA when the population canonical directions are sufficiently sparse. The effective dimension is controlled by the sparsity levels s_u and s_v, rather than by the full ambient dimensions p and m. In the rank-one case, the consistency condition has the familiar form

$$\frac{s_u \log p}{n\lambda^2} \to 0, \qquad \frac{s_v \log m}{n\lambda^2} \to 0.$$

Thus, even when p and m are much larger than n, the canonical directions can still be consistently estimated as long as the true directions are sparse, the canonical correlation signal is not too weak, and the sparse complexity terms vanish.

2.3.2 Variable Selection Consistency

Beyond estimating the directions themselves, one may ask whether SCCA can recover the correct support:

$$\widehat{S}_a = \{j : \hat{a}_{1j} \neq 0\}, \qquad \widehat{S}_b = \{j : \hat{b}_{1j} \neq 0\}.$$

This is a stronger requirement than estimation consistency. Small ℓ_2 error does not automatically imply exact support recovery unless the nonzero coefficients are separated from zero.

Theorem 2 (Support recovery in sparse CCA; Laha and Mukherjee, 2022)**.** *For high-dimensional sparse CCA, exact support recovery is possible only under additional sample-size and signal-strength conditions. In particular, if the sparsity levels are not too large relative to n and $p_1 + p_2$, and the minimum nonzero entries of the canonical vectors are sufficiently large, then consistent support recovery is achievable. Conversely, when the sparsity level is too high or the minimum signal strength is too small, exact support recovery is information-theoretically impossible.*

Thus, variable selection consistency should not be presented as a direct consequence of the usual SCCA consistency theorem. It requires separate support-recovery theory. Laha and Mukherjee (2022) give such a theory: they define successful support recovery as exact recovery with probability tending to one, identify sparsity regimes where recovery is possible, and show that sufficiently large minimum nonzero entries are necessary for asymptotically consistent support recovery.

2.3.3 Minimax Lower Bounds

Another theoretical question is whether the convergence rates achieved by SCCA estimators are optimal. Gao, Ma, Ren, and Zhou (2015) study sparse CCA from a decision-theoretic perspective and establish minimax lower and upper bounds for estimating the leading sparse canonical directions.

Theorem 3 (Minimax lower bound; Gao, Ma, Ren, and Zhou, 2015)**.** *Let $\mathbf{U}_1$ and $\mathbf{V}_1$ denote the leading sparse canonical direction matrices. Under the sparse CCA model of Gao et al. (2015), the minimax risk for estimating $\mathbf{U}_1\mathbf{V}_1^\top$ satisfies*

$$\inf_{\widehat{\mathbf{U}}_1,\widehat{\mathbf{V}}_1} \sup_{\Sigma\in\mathcal{F}_q} \mathbb{E}_\Sigma \left\|\widehat{\mathbf{U}}_1\widehat{\mathbf{V}}_1^\top - \mathbf{U}_1\mathbf{V}_1^\top\right\|_F^2 \geq c\,\varepsilon_n^2 \wedge c_0,$$

where

$$\varepsilon_n^2 = \frac{1}{n\lambda^2}\left[r(k_q^u + k_q^v) + k_q^u \log\left(\frac{\mathrm{e}p_1}{k_q^u}\right) + k_q^v \log\left(\frac{\mathrm{e}p_2}{k_q^v}\right)\right].$$

For exactly sparse leading directions with rank $r = 1$, this rate reduces heuristically to

$$\frac{s_1 \log(\mathrm{e}p_1/s_1) + s_2 \log(\mathrm{e}p_2/s_2)}{n\lambda^2}.$$

This result belongs in the minimax subsection, not the basic consistency subsection: it is about the optimal risk rate, not merely convergence in probability. Gao et al. define this rate in their equation (15), prove a matching upper bound in Theorem 1, and give a minimax lower bound in Theorem 3.

2.3.4 Finite-Sample Guarantees

The same Gao et al. (2015) result also yields a finite-sample high-probability guarantee. Under their assumptions, the sparse CCA estimator satisfies

$$\left\|\widehat{\mathbf{U}}_1\widehat{\mathbf{V}}_1^\top - \mathbf{U}_1\mathbf{V}_1^\top\right\|_F^2 \leq C\varepsilon_n^2$$

with probability at least

$$1 - \exp\left[-C'\left(k_q^u + \log\frac{\mathrm{e}p_1}{k_q^u}\right)\right] - \exp\left[-C'\left(k_q^v + \log\frac{\mathrm{e}p_2}{k_q^v}\right)\right].$$

This is a finite-sample statement because it gives an explicit probability bound for fixed n, p_1, p_2, s_1, s_2, rather than only an asymptotic convergence statement.

2.4 Connections to Other Methods

2.4.1 Sparse PCA

Sparse PCA (Zou, Hastie, and Tibshirani 2006) seeks sparse principal component loading vectors. The relationship to SCCA is direct: if one sets $\mathbf{Y} = \mathbf{X}$ in (2), then $\mathbf{M} = \mathbf{X}^\top \mathbf{X}$ and the PMD problem reduces to finding sparse eigenvectors of the sample covariance matrix. This observation motivates a unified algorithmic framework for sparse matrix decompositions (Witten, Tibshirani, and Hastie 2009).

When $\mathbf{M} = \mathbf{X}^\top \mathbf{X}$ is symmetric and positive semidefinite, the PMD(ℓ_1, ℓ_1) formulation is closely related to sparse PCA. In this setting, the bilinear objective

$$\max_{\|\mathbf{u}\|_2 \le 1,\ \|\mathbf{v}\|_2 \le 1,\ \|\mathbf{u}\|_1 \le c_1,\ \|\mathbf{v}\|_1 \le c_2} \mathbf{u}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v}$$

can be viewed as a sparse low-rank approximation to the covariance matrix. Under a symmetric loading interpretation, this leads to the sparse eigenvector-type problem

$$\max_{\|\mathbf{v}\|_2 \le 1,\ \|\mathbf{v}\|_1 \le c} \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v},$$

which is a convex-constraint relaxation of the sparse principal component problem. The same soft-thresholding updates used in SCCA therefore reappear in sparse PCA, with the left and right loading vectors corresponding to the same underlying sparse direction.

2.4.2 Reduced-Rank Regression

Reduced-rank regression (RRR; Anderson, 1951; Izenman, 1975; Reinsel and Velu, 1998) studies the multivariate regression model

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E},$$

where $\mathbf{Y} \in \mathbb{R}^{n \times p_2}$, $\mathbf{X} \in \mathbb{R}^{n \times p_1}$, and the coefficient matrix $\mathbf{B} \in \mathbb{R}^{p_1 \times p_2}$ is constrained to have low rank:

$$\operatorname{rank}(\mathbf{B}) \le r.$$

This rank constraint forces the fitted mean $\mathbb{E}(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\mathbf{B}$ to depend on $\mathbf{X}$ through only a small number of latent linear combinations. In this sense, RRR and CCA are closely related: both methods seek low-dimensional linear structure linking two sets of variables.

However, RRR is not identical to ordinary CCA. CCA is a symmetric association method: it finds linear combinations of $\mathbf{X}$ and $\mathbf{Y}$ with maximal correlation. In contrast, RRR is a conditional mean model for predicting $\mathbf{Y}$ from $\mathbf{X}$. Thus, the two methods answer different questions. CCA asks which directions in $\mathbf{X}$ and $\mathbf{Y}$ are most strongly associated, whereas RRR asks which low-dimensional subspace of predictors is most useful for explaining or predicting the response matrix.

The connection between the two methods appears through generalized eigenvalue problems involving the covariance structure of $\mathbf{X}$ and $\mathbf{Y}$. At the population level, the multivariate least-squares coefficient is

$$\mathbf{B}_{\text{OLS}} = \mathbf{\Sigma}_{XX}^{-1} \mathbf{\Sigma}_{XY},$$

assuming $\mathbf{\Sigma}_{XX}$ is nonsingular. RRR then replaces this full-rank coefficient matrix by a rank-r approximation, but the approximation is taken with respect to the response covariance structure rather than by simply truncating the ordinary Euclidean singular value decomposition of $\mathbf{B}_{\text{OLS}}$. Therefore, a formula such as

$$\widehat{\mathbf{B}} = \widehat{\mathbf{B}}_{\text{OLS}} \mathbf{V}_r \mathbf{V}_r^\top$$

should be understood only as a schematic description, not as the general RRR estimator.

A more precise way to state the relationship is that RRR can be expressed through canonical analysis of the fitted response space. The rank constraint selects a small number of response directions that are most predictable from $\mathbf{X}$, while CCA selects pairs of predictor and response directions that are maximally correlated after normalizing by the within-set covariance matrices. Hence, RRR is closely connected to CCA, but the canonical coefficient vectors are not, in general, simply proportional to the regression coefficient vectors.

Sparse reduced-rank regression extends this idea by imposing sparsity on the coefficient structure in addition to the rank constraint. For example, Chen and Huang (2012) study sparse reduced-rank regression by combining low-rank structure with variable selection. This is conceptually parallel to SCCA: both methods seek interpretable low-dimensional relationships between two high-dimensional data matrices. The difference is that sparse RRR is prediction-oriented and asymmetric, whereas SCCA is association-oriented and symmetric.

2.4.3 Graphical Lasso and Partial Correlations

Graphical-lasso-type methods provide a conditional-dependence perspective on the relationship between two sets of variables. Let

$$\mathbf{Z} = \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix}, \qquad \mathbf{\Sigma} = \begin{pmatrix} \mathbf{\Sigma}_{XX} & \mathbf{\Sigma}_{XY} \\ \mathbf{\Sigma}_{YX} & \mathbf{\Sigma}_{YY} \end{pmatrix},$$

and let

$$\mathbf{\Omega} = \mathbf{\Sigma}^{-1} = \begin{pmatrix} \mathbf{\Omega}_{XX} & \mathbf{\Omega}_{XY} \\ \mathbf{\Omega}_{YX} & \mathbf{\Omega}_{YY} \end{pmatrix}$$

denote the precision matrix of the joint vector $(\mathbf{X}^\top, \mathbf{Y}^\top)^\top$.

Under a Gaussian graphical model, zeros in the precision matrix correspond to conditional independences between variables, conditional on all remaining variables. This idea goes back to covariance selection in Dempster (1972) and is treated systematically in Lauritzen (1996). The graphical lasso estimates a sparse precision matrix by solving the penalized likelihood problem

$$\widehat{\mathbf{\Omega}} = \arg\max_{\mathbf{\Omega} \succ 0} \left\{ \log\det(\mathbf{\Omega}) - \operatorname{tr}(\widehat{\mathbf{\Sigma}}\mathbf{\Omega}) - \lambda \|\mathbf{\Omega}\|_1 \right\},$$

where $\widehat{\mathbf{\Sigma}}$ is the sample covariance matrix and $\lambda > 0$ controls the sparsity level; this formulation is commonly associated with the graphical lasso of Friedman, Hastie, and Tibshirani (2008). Equivalently, this estimator can be viewed as an ℓ_1-regularized maximum-likelihood estimator for the inverse covariance matrix.

The entries of the precision matrix are closely related to partial correlations, but they are not themselves partial correlations. For variables i and j, the partial correlation after conditioning on all remaining variables

is

$$\rho_{ij|-ij} = -\frac{\Omega_{ij}}{\sqrt{\Omega_{ii}\Omega_{jj}}}.$$

Thus, the off-diagonal block $\mathbf{\Omega}_{XY}$ encodes conditional cross-group dependence between $\mathbf{X}$ and $\mathbf{Y}$ after adjusting for all other variables.

This perspective is complementary to SCCA. Sparse CCA searches for sparse linear combinations of $\mathbf{X}$ and $\mathbf{Y}$ with high marginal correlation, while graphical-lasso-type methods identify sparse conditional dependence relationships among the individual variables. In other words, SCCA is association-oriented and low-dimensional, whereas graphical lasso is conditional-dependence-oriented and graph-structured. Both methods impose sparsity, but they regularize different objects: SCCA regularizes canonical loading vectors, while graphical lasso regularizes the precision matrix.

2.4.4 Partial Least Squares

Partial least squares (PLS) is another method for finding low-dimensional relationships between two data matrices; see Rosipal and Krämer (2006) for an overview. Like CCA, PLS constructs linear combinations of two sets of variables, but the optimization criterion is different. PLS seeks directions $\mathbf{a}$ and $\mathbf{b}$ that maximize covariance,

$$\mathrm{Cov}(\mathbf{Xa}, \mathbf{Yb}),$$

often under Euclidean norm constraints such as

$$\|\mathbf{a}\|_2 \leq 1, \qquad \|\mathbf{b}\|_2 \leq 1.$$

In contrast, classical CCA maximizes correlation,

$$\mathrm{Corr}(\mathbf{Xa}, \mathbf{Yb}),$$

or equivalently maximizes covariance after normalizing by the within-set variances

$$\mathbf{a}^\top \mathbf{\Sigma}_{XX}\mathbf{a} = 1, \qquad \mathbf{b}^\top \mathbf{\Sigma}_{YY}\mathbf{b} = 1.$$

This distinction is important. Because PLS maximizes covariance rather than correlation, it is generally sensitive to the scaling of the original variables unless the variables are standardized beforehand. CCA, by contrast, is invariant to separate rescalings of the variables, provided the within-set covariance matrices are nonsingular. On the other hand, PLS avoids directly inverting $\widehat{\mathbf{\Sigma}}_{XX}$ and $\widehat{\mathbf{\Sigma}}_{YY}$, which can make it numerically more stable than classical CCA in high-dimensional or highly collinear settings.

2.5 Limitations and Directions for Future Work

2.5.1 Statistical Limitations

Despite its usefulness, Sparse CCA still has several important limitations.

1. **Non-convexity.** The Witten–Tibshirani–Hastie PMD formulation is bilinear in $(\mathbf{a}, \mathbf{b})$:

$$\max_{\mathbf{a},\mathbf{b}} \mathbf{a}^\top \widehat{\mathbf{\Sigma}}_{XY} \mathbf{b} \quad \text{subject to sparsity and norm constraints.}$$

   Although the objective is convex in one block when the other block is fixed, it is not jointly convex in $(\mathbf{a}, \mathbf{b})$. Therefore, alternating updates should be interpreted as a coordinate-ascent procedure: the objective is nondecreasing along the iterations, but the algorithm is generally only expected to reach a coordinate-wise stationary point, not necessarily a global maximizer. In practice, multiple initializations are recommended.

2. **Tuning parameter selection.** The sparsity parameters c_1 and c_2 control the number of variables selected in the two canonical loading vectors. Choosing them jointly is computationally expensive, because it usually requires searching over a two-dimensional grid. Witten, Tibshirani, and Hastie (2009) propose a permutation-based tuning procedure, but the selected model can still depend on the grid, the resampling scheme, and the desired tradeoff between sparsity and canonical correlation. Cross-validation, permutation-based selection, and stability-based selection are all reasonable approaches, but there is no universally accepted tuning rule for SCCA.

3. **Sequential deflation for higher-order pairs.** Higher-order sparse canonical pairs are commonly obtained by deflating the cross-covariance matrix after estimating the leading pair. This is algorithmically convenient, but it is not equivalent to the classical orthogonality structure of ordinary CCA. Errors in the first estimated pair may propagate to later pairs, and the resulting sparse canonical vectors are not automatically guaranteed to satisfy the same orthogonality relations as classical canonical vectors. Joint estimation of several sparse canonical pairs may reduce this sensitivity, but it is computationally more difficult.

4. **Inference after selection.** Although SCCA theory provides results on point estimation, support recovery, and minimax rates, valid post-selection inference for entries of $\hat{\mathbf{a}}_k$ and $\hat{\mathbf{b}}_k$ remains difficult. The main obstacle is that SCCA estimators come from a nonlinear, non-smooth, and generally non-convex bilinear optimization problem.

   Debiased or desparsified lasso methods provide a useful analogy. In high-dimensional regression, Zhang and Zhang (2014) and van de Geer et al. (2014) correct the biased lasso estimator by adding a score-type term involving an approximate inverse of the Gram matrix; under suitable sparsity and regularity conditions, the corrected estimator is asymptotically normal and can be used for confidence intervals and tests. However, these results apply to regression-type models, not directly to SCCA. Extending such post-selection inference ideas to SCCA is nontrivial because SCCA has two coupled loading vectors, normalization constraints, sign indeterminacy, and a bilinear objective.

2.5.2 Methodological Extensions

Several directions remain important for future work.

1. **Nonlinear SCCA.** Kernel CCA extends classical CCA to nonlinear relationships by mapping the two data views into reproducing kernel Hilbert spaces (Hardoon, Szedmak, and Shawe-Taylor, 2004). A natural extension is to combine nonlinear feature maps with sparsity or feature selection, so that the resulting method can capture nonlinear association while remaining interpretable. However, such methods introduce additional tuning choices, such as kernel parameters and sparsity penalties, and may be more difficult to analyze theoretically.

2. **Multiview and generalized CCA.** Many modern applications involve more than two data blocks, such as multiple omics platforms, multiple imaging modalities, or several market segments. Multiset CCA and generalized CCA extend the CCA framework beyond two views (Carroll, 1968; Kettenring, 1971; Rupnik and Shawe-Taylor, 2010). Sparse versions of generalized CCA are a natural continuation of SCCA, because they aim to identify a small number of relevant variables across multiple views rather than only two.

3. **Temporal and dependent data.** The financial-return analysis in Part I shows strong volatility clustering and time-varying covariance structure. This does not necessarily imply strong serial correlation in the raw returns, but it does indicate that the i.i.d. assumptions behind standard CCA and SCCA theory are questionable for financial data. Future work could combine SCCA with time-series models, for example by first filtering returns using GARCH-type models, by estimating canonical directions from standardized residuals, or by allowing the canonical vectors and covariance matrices to evolve over time.

4. **Large-scale and streaming CCA.** For very large data sets, classical CCA can be computationally expensive because it requires estimating and decomposing large covariance matrices. Scalable and stochastic CCA algorithms, such as those of Ma, Lu, and Foster (2015), address the computational side of this problem. However, these methods are not themselves sparse CCA procedures. A useful future direction is to develop streaming or online SCCA algorithms that update sparse canonical vectors as new observations arrive, while preserving both computational efficiency and interpretability.

2.6 Summary (Review Project)

We have reviewed Sparse Canonical Correlation Analysis (SCCA), with particular emphasis on the penalised matrix decomposition formulation of Witten, Tibshirani, and Hastie (2009), its algorithmic structure, its theoretical properties, and its relationship to other multivariate methods.

The main points are as follows.

1. **Formulation.** Classical CCA seeks linear combinations of two random vectors that are maximally correlated, but the sample version becomes unstable or ill-defined when the within-view covariance matrices are singular or poorly conditioned. SCCA addresses this problem by adding sparsity constraints or penalties to the canonical coefficient vectors. In the Witten–Tibshirani–Hastie formulation, this leads to the PMD(ℓ_1, ℓ_1) problem, which is solved by alternating updates with soft-thresholding. The resulting canonical vectors are sparse and therefore more interpretable in high-dimensional settings.

2. **Theory.** The theoretical literature shows that sparsity is essential for consistent estimation in high-dimensional CCA. Under suitable structural and regularity assumptions on the population covariance matrices and the true canonical directions, sparse CCA estimators can consistently recover the leading canonical subspaces. Minimax results, such as those of Gao, Ma, and Zhou (2015), quantify the best possible estimation error over sparse parameter spaces, while variable-selection results, such as those of Mai and Zhang (2019), require additional signal-strength and design conditions. Thus, sparsity helps both statistically and interpretively, but the precise guarantee depends on the estimator, the loss function, and the assumptions imposed.

3. **Connections.** SCCA is closely related to several other multivariate methods. It connects to sparse PCA through the PMD framework, since sparse PCA can be viewed as a one-view analogue of the same penalised low-rank decomposition idea. It is also related to reduced-rank regression through the classical connection between CCA and multivariate regression. Graphical-lasso methods are complementary rather than equivalent: they estimate sparse conditional-dependence structure through precision matrices, whereas SCCA estimates sparse cross-view linear associations. Finally, SCCA differs from partial least squares because CCA maximises correlation after within-view normalisation, whereas PLS maximises covariance and is therefore more sensitive to marginal scale.

4. **Limitations.** Despite its usefulness, SCCA has several limitations. The optimisation problem is generally non-convex, so the computed solution may depend on the algorithm and initialization. Tuning sparsity parameters is also difficult, especially when the goal is stable variable selection rather than merely high canonical correlation. Higher-order canonical pairs require deflation, but sequential deflation can introduce bias and make later components harder to interpret. Inference is another open challenge: valid confidence intervals or hypothesis tests for individual selected coefficients are difficult because selection, non-convexity, and high dimensionality interact.

5. **Connection to Part I.** The empirical analysis in Part I illustrates why SCCA is useful. At the sector-ETF level, the problem is low-dimensional, so classical CCA is numerically stable and already interpretable. At the individual-stock level, however, the IT–Health Care analysis has $p_1 = 67$ and $p_2 = 55$, so classical CCA is much more vulnerable to high-dimensional upward bias. The permutation null in Part I agrees closely with the Bai–Miao–Pan high-dimensional bias prediction, showing that a large sample canonical correlation can arise partly from dimensional effects. The sparse CCA fit at

$(c_1, c_2) = (0.3, 0.3)$ then produces a much smaller and more interpretable active set while preserving most of the leading canonical correlation. This empirical pattern is consistent with the main message of the review: sparsity is not only a computational device, but also a way to extract interpretable cross-view structure from high-dimensional data.

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